THE DARK ROOM IN THE REWARD CHANNEL: DENSE PREDICTION REWARDS COLLAPSE GRPO-TRAINED LLM AGENTS, AND THE CHANNEL, NOT THE CONTENT, DECIDES WHAT WORKS

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ABSTRACT

Sparse success rewards make long-horizon LLM agents slow to train; the common remedy is dense per-step supervision: reward the policy for predicting its next observation, which looks provably safe under potential-based shaping. Published prediction-reward and auxiliary-loss variants report successes and instabilities; no controlled account establishes when the reward channel is dangerous or where auxiliary gains come from. We give that account: 74 preregistered arms dissect one fixed prediction signal under group-normalized RL (GRPO) across ALFWorld, WebShop, a synthetic POMDP, and Qwen3-1.7B/4B/8B, varying only the delivery mechanism. (1) Every run sustaining this difference-form reward under untouched std normalization (no filtering, dynamic-sampling, or decoupling mitigations) collapses: eleven runs across scales, coefficients, group sizes, and groupings (the floor-bound synthetic environment stalls instead); ALFWorld runs end in a degenerate absorbing state (prediction accuracy $\to$1.0, success $\to$0): the optimizer builds the "dark room". One line of algebra explains it: inside all-fail groups, z-scoring cancels the shaping coefficient; removing only std normalization restores baseline parity. (2) A signal's danger is set by its within-group *variance trajectory*, with hackability as a second axis; it retrodicts every reward-channel collapse and survives preregistered prospective tests. (3) The same signal as an auxiliary teacher-forced loss does no harm on ALFWorld at 4B, but the gain is not the signal's: content-free placebos as a class match or beat gold at both matched seeds (seed 0: placebo mean 78.8 vs. 68.6; seed 42: single endpoint 67.9 vs. 57.9); the auxiliary update is the regularizer. (4) At 8B the recipe turns bistable: gold full-weight locks two of three seeds; every content-free or reduced-weight arm stays healthy. No ALFWorld or WebShop reward-channel variant measurably beats its matched-normalization baseline and no gold signal measurably outperforms its content-free placebo: the delivery channel, not the content, decides; which channel is safe is regime-dependent.

1 INTRODUCTION

Long-horizon LLM agents trained with sparse success rewards learn slowly. A natural remedy is dense per-step supervision: ask the policy to predict its next observation and reward it for being right. Some form of this recipe is on the menu wherever agent RL meets sparse rewards, which today means most GRPO-family agent training. We show the remedy backfires catastrophically under group-normalized RL (Shao et al., 2024); Figure 1 maps the setup, the failure cascade, and the two deliveries that survive our controls. The *dark room problem* (Friston et al., 2012) asks why a surprise-minimizing agent does not retreat to a fully predictable corner; long a philosophical puzzle for predictive-processing theories (Baltieri & Buckley, 2019; Seth et al., 2020), it is a concrete engineering outcome in our experiments: at every scale, group size, and fixed shaping coefficient we tested, the optimizer builds the dark room automatically (timed withdrawal inside the early gain window is the one schedule that escapes, §5.2). Curiosity-driven agents chase prediction

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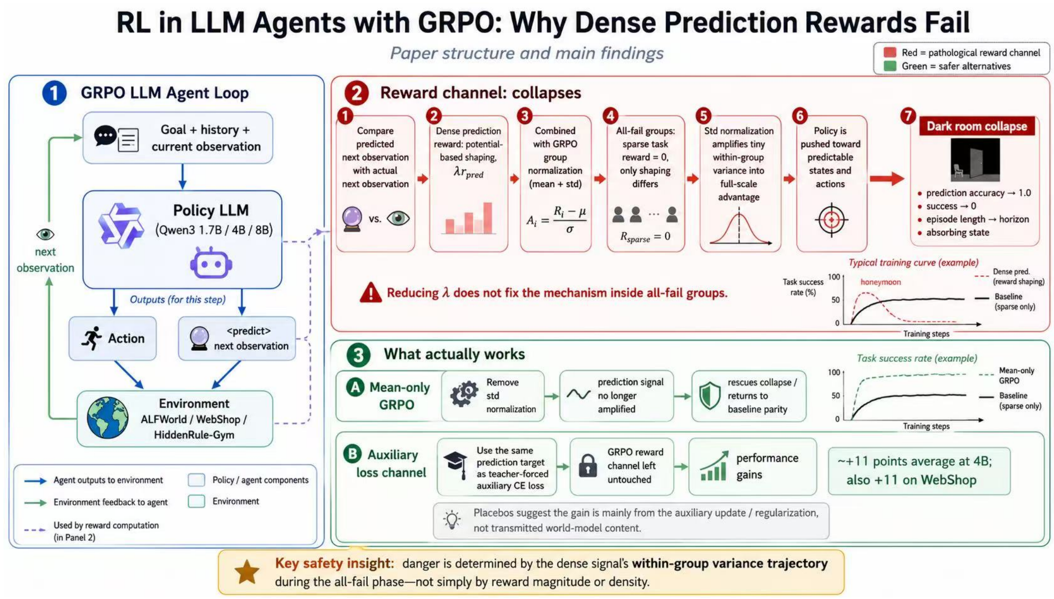

Figure 1: Overview of the study. *Panel 1*: the step-independent GRPO agent loop with a verifiable `<predict>` block scored against the next observation (§3.1). *Panel 2*: the failure cascade in the reward channel: in all-fail groups the ε-scale shaping term is the only within-group variation, std normalization amplifies it to full-scale advantages, and the policy converges to a dark-room absorbing state; reducing λ does not change the update (§3.2, §5). *Panel 3*: the two deliveries that survive our controls: mean-only normalization (rescue to baseline parity) and the auxiliary-CE loss channel, whose gain the placebo ladder attributes to the update itself rather than the world-model content (§5.4). Annotated gains are rounded summaries: the 4B auxiliary gain measures $+14.2$ training-caliber / $+15.4$ formal on three-seed means (§6.4); the WebShop $+11.0$ is training-caliber and seed-split (formal $+3.0$, statistically level; §6); and the loss channel's safety is scale-qualified by the 8B bistability (§6). The boxed insight's "all-fail phase" is the sparse-success route into the variance-dominated window; a continuous-score second route exists (§3.3).

error and get trapped by noisy TVs (Burda et al., 2019); ours chase prediction *accuracy* and get trapped by their own predictability: the same failure reached from opposite directions. The failure is puzzling because the signal meets every safety standard the shaping literature asks for. In our step-independent multi-turn setting (Feng et al., 2025), the policy emits a verifiable `<predict>` block each step, rule-scored against the actual next observation with no judge model; the score enters as a difference-form per-step reward, superficially shaped like potential-based shaping (PBRS), whose episode sum telescopes so return-level distortion is provably at most λ. Yet every run that sustains the difference-form reward under untouched std normalization in our sweep collapses. On ALFWorld the natural history is stereotyped: at 4B and 8B an early gain phase (a "honeymoon," absent at 1.7B) whose peaks rise with scale though never above the arm's formal baseline, then a turn, then an absorbing state from which no intervention we tested recovers; on WebShop the honeymoon is replaced by instant suppression or baseline parity (§6). The root cause lives one level below returns, in the advantage estimator. Sparse success makes all-fail groups common, and inside such a group the only within-group variation is the ε-scale shaping term. GRPO's z-scoring divides by the group's own standard deviation, so the normalized advantage is *invariant to the shaping coefficient* (Proposition 1): bounded rewards become full-scale pressure, and, while the shaping term dominates the pool's residual spread ($\lambda\sigma_s \gg \max(\sigma_p, \epsilon)$, §3.2), no reduction of the coefficient short of zero changes the update. A three-point λ-sweep confirms this (amplified advantage scale $\pm 7.3/\pm 9.1/\pm 7.2$ across a tenfold range; every arm collapses), and a single-factor ablation completes the causal case: removing only the std normalizer turns 0% into baseline parity, and a knock-out matrix (remove the signal, the amplifier, or the all-fail groups) shows each ingredient is necessary. The mechanism generalizes into the paper's organizing principle, a *variance-trajectory criterion* (endpoint case formalized in Proposition 2): what z-scoring amplifies is a dense signal's within-group variance, so a signal's danger is set by whether that variance survives the phase when it dominates the group advantage (all-fail domination is the sparse-success route into that phase; a continuous-score easy regime is a second route), not by magnitude or density. The criterion retrodicts every reward-channel collapse in this paper and every null but one (anchor-QA, where it over-warns: variance persisted yet the formal endpoint is level with baseline, §3.3), makes preregistered prospective predictions (full SHA256 digests committed to a public version-controlled file before the runs completed, shipped with the code release), and is compatible with published reward-channel successes. An early-warning rule built on it detects all three held-out reward-channel collapses 13–73 steps in advance, with at most one false alarm among the eleven training-healthy arms (a set distinct from the abstract's eleven collapsing runs; the flagged arm was itself later revised below baseline by the formal evaluation, blurring even that alarm's status).

The criterion says when the reward channel is dangerous; a controlled signal-delivery matrix says what to do instead, and why the popular explanation of the fix is wrong. Holding signal, model, environment, and budget fixed and varying only the consumption mechanism, every reward-channel variant on ALFWorld and WebShop is at best neutral. The same signal as an auxiliary teacher-forced loss does no harm, and what gain appears is not the signal's: under the identical recipe a placebo ladder (shuffled gold, random-vocabulary, random-token) matches or beats gold at both matched seeds (the placebo mean 78.8 vs. gold 68.6 at seed 0; the single s42 placebo endpoint 67.9 vs. 57.9), a compute-matched no-target control shows no gain (48.6%), and the random-vocabulary placebo tops the entire table; a cross-normalization convergence reinforces the reading descriptively (fixing the normalizer alone, with no signal at all, reaches the same level, 57.9% formal). Each placebo family carries its own three-seed replication (at training caliber five of six seeds above the baseline band, the sixth inside it; none below, none collapsing), so the class-level ordering does not rest on a single run. What works is the auxiliary CE pass itself, acting as a regularizer; the world-model content plays no measurable role, which bears awkwardly on the auxiliary-world-model literature, whose gains may partly rest on the same effect.

Both findings generalize within boundaries we map in §6. On WebShop the reward channel fails again, instantly in the easy regime and through the familiar collapse-after-parity shape in the hard regime, while the auxiliary channel is seed-split in the easy regime (single-seed-pair $+11.0/-9.9$; formal $+3.0$ statistically level and -14.0; the hard-regime cell is untested). At 8B the auxiliary recipe itself becomes *bistable*: gold full-weight training locks two of three seeds into an absorbing state and sends the third to the study's best gold-signal result, while every content-free or reduced-weight arm stays healthy. The randomness has moved from collapse timing to collapse occurrence, and content-inertness, intact on the gain side, reverses on the risk side.

We are not proposing that world-model objectives be added to agents; we report what happens when they are, so the recipe under study is not itself a contribution. Our contributions are:

1. **Mechanism and causal localization of a catastrophic failure.** Three-scale collapse phenomenology; a one-line invariance result (Proposition 1) with three-point confirmation; a single-factor rescue and knock-out matrix; a behavioral characterization of the absorbing state; irreversibility across scheduling, coefficient scale, and post-hoc structural interventions (§3, §5).
2. **A signal-side safety criterion with prospective validation.** The variance-trajectory criterion (endpoint case: Proposition 2), its retrodictions (complete on collapses; one over-warning on nulls), preregistered prospective tests, a validated early-warning signature, and a compatibility check against published successes (§3.3, §5).
3. **The channel effect, fully attributed.** A fourteen-arm delivery matrix (the same signal across every signal-bearing arm, plus matched controls) and placebo ladder attributing the auxiliary-loss gain to the update itself, with seed replication and a controlled challenge to world-modeling attributions in prior work (§5.4).
4. **Scale and environment boundaries of both results.** Cross-environment replication of the failure and the content-free attribution on WebShop (where the auxiliary gain is seed-split), a regime-dependent failure-form taxonomy (instant suppression vs. collapse-after-parity), and the 8B bistability: same recipe, seed-determined locked collapse or the study's best gold-signal gain (§6).

2 Related Work

Group-normalized RL and its pathologies. GRPO (Shao et al., 2024; DeepSeek-AI, 2025) normalizes group returns by mean and std; Dr.GRPO (Liu et al., 2025) identified the std term's biases, and we adopt its mean-only variant as an *instrument*, contributing the causal demonstration that in long-horizon sparse-success agents the std term alone separates catastrophic from benign. Our results delimit where the std-as-adaptive-gradient defense (Ge et al., 2026) breaks: phases where an ε-scale shaping term supplies the dominant within-group variance, with sparse-success all-fail groups as the extreme case. A distinct line treats degenerate groups as zero-signal dead zones to prune or reweight (Xu & Ding, 2025; Zhang et al., 2026); our setting inverts that premise (all-fail groups become *full-signal* zones, Proposition 1), and remedies that merely skip them remove the amplifier's fuel without the amplifier, rescue only partially (-9.9pt, §5.3), and can mask the mechanism. Proposition 1 is, to our knowledge, the first formalization of this amplification for the sparse-success all-fail structure of multi-turn agents.

Dense signals and shaping. Prediction-*error* seeking falls to the noisy-TV trap (Pathak et al., 2017; Burda et al., 2019); our failure is the mirror image, prediction-*accuracy* seeking parking in predictable corners (Friston et al., 2012). Ng-style invariance (Ng et al., 1999) assumes a policy-independent potential; the known ways it breaks (Devlin & Kudenko, 2012; Behboudian et al., 2021) never consider group normalization, where our damage concentrates. The standard hacking mitigation is the progress principle (Setlur et al., 2025; Hou et al., 2025); our Δacc arm is its preregistered test, and its failure sharpens the boundedness principle of Fu et al. (2025): return-level boundedness does not survive group z-scoring at the step level.

Auxiliary objectives and world modeling. Loss-channel world-model signals have a consistent track record (Jaderberg et al., 2017; Shrivastava et al., 2026; Lu et al., 2026); the reward channel is not uniformly hostile (RWML (Yu et al., 2026), VAGEN (Wang et al., 2025b)), and our criterion must be, and is, consistent with those successes (§3.3). What this literature lacks is a controlled comparison holding the signal fixed while varying only the consumption mechanism; our matrix supplies it (§5.4), and relative to ECHO (Shrivastava et al., 2026) we contribute the attribution. Our 8B bistability is not generic seed variance (Dodge et al., 2020) rebranded (§6). Extended related work, including grouping-scheme successors, qualitative precedents of the amplification, and scale-instability context, is in Appendix G.

3 Method: Signal, Amplification, and the Variance Criterion

Three modules organize the paper's apparatus: the verifiable prediction signal and how it enters the reward (§3.1); the amplification algebra that turns a bounded signal into full-scale pressure (§3.2); and the variance-trajectory criterion that generalizes it (§3.3).

3.1 Setting: verifiable prediction as a shaping reward

Agent framework. Step-independent multi-turn rollout (Feng et al., 2025): each step's LLM input is built from the current observation plus a bounded history summary, keeping context length near-constant over 15–50 step horizons (per-environment, Appendix C). Training uses GRPO (Shao et al., 2024) with grouped environments (group size 4) and terminal task reward (ALFWorld: sparse success 10 / failure 0; WebShop replaces the binary outcome with a continuous terminal matching score in $[0, 1]$, the "continuous scores" of §3.3), plus an invalid-action penalty.

Prediction-sufficiency (PS) signal. Prompts request a `<predict>` block over a task-agnostic feature schema Φ (location, objects-visible boolean, receptacle state; an open-set visible-objects F1 is logged but never rewarded). The block is rule-parsed and verified against the *next* environment observation; no judge model is involved. The per-step reward is the difference form $r_{\text{pred}}(t) = \Phi_t - \Phi_{t-1}$ scaled by λ=0.1, injected on the last response token of each step sample.

3.2 Bounded returns, unbounded advantages

Let a group contain G trajectories with returns R_i; GRPO's normalized advantage and our per-step shaping reward are

$$\hat{A}_i \;=\; \frac{R_i - \bar{R}}{\sigma_R + \epsilon}, \qquad r_{\text{pred}}(t) \;=\; \lambda \left(\Phi_t - \Phi_{t-1}\right), \tag{1}$$

where $\Phi_t \in [0, 1]$ scores the policy's own prediction against the realized next observation. Because the shaping telescopes,

$$\sum_t r_{\text{pred}}(t) \;=\; \lambda \left(\Phi_T - \Phi_0\right), \qquad \Big|\sum_t r_{\text{pred}}(t)\Big| \;\le\; \lambda, \tag{2}$$

episode returns are distorted by at most λ: at the *return* level the arithmetic bound delivers the safety that PBRS rhetoric suggests. The formal PBRS invariance theorem (Ng et al., 1999) does not apply, however: Φ scores the policy's *own* predictions, a policy-dependent potential, exactly the class for which invariance is known to break (§2). The compliance is superficial, and the failure enters one level down.

The normalization unit. One implementation fact is load-bearing for everything that follows. Each of a trajectory's T steps is a separate sample in the update batch, and the GRPO estimator computes the group mean and std over *all step-samples sharing a group ID* (its cross-step default), not over the G trajectory returns: the pool has $n = G{\cdot}T \approx 4{\times}50 \le 200$ members, each scored by its own per-step reward, and each sample's z-score is broadcast to its response tokens. The propositions below use trajectory notation for readability, but the algebra is unit-agnostic and the unit matters quantitatively: a z-score over n pooled values is bounded by $(n-1)/\sqrt{n}$, i.e. 1.5 at $n{=}4$ but $\approx$14 at $n{=}200$. The advantage magnitudes we measure ($\pm$5–9, §5) are unreachable under trajectory pooling and sit inside the step-sample bound; the fingerprint itself identifies the unit. The unit also dissolves an apparent tension with Eq. 2: telescoping bounds the return-level distortion, but the normalizer never sees that sum; it divides by the pooled spread of the *per-step* values $\lambda(\Phi_t - \Phi_{t-1})$, which keeps fluctuating as long as the policy visits varied states. A shaping term can therefore be return-bounded and advantage-dominant at once. One more ε-scale term shares the pool: the invalid-action penalty (0.1, the same order as λ). Proposition 1 states both limits; the λ-sweep alone cannot apportion the pooled spread, and the causal apportioning (every baseline arm carries the same penalty under the same normalizer and the same early all-fail exposure and never collapses; the terminal behavior is prediction-optimal, the imprint of shaping pressure during the approach) assigns the operative pressure to the shaping component. The full identifiability discussion, including the implementable-but-unlogged spread monitor, is in Appendix N.

Proposition 1 (λ-invariance in all-fail groups). *In a group where every trajectory fails the task, write $s_i = \sum_t (\Phi_t - \Phi_{t-1})_i$ for the accumulated shaping potential and p_i for the accumulated invalid-action penalty, so $R_i = C + \lambda s_i - p_i$ with C the shared task return. If the shaping term dominates the within-group spread ($\lambda\sigma_s \gg \max(\sigma_p, \epsilon)$),*

$$\hat{A}_i \;=\; \frac{\lambda\,(s_i - \bar{s}) - (p_i - \bar{p})}{\sigma_{\lambda s - p} + \epsilon} \;\longrightarrow\; \frac{s_i - \bar{s}}{\sigma_s}, \tag{3}$$

which is independent of λ. In the opposite limit ($\sigma_p \gg \max(\lambda\sigma_s, \epsilon)$) the advantage tends to $-(p_i - \bar{p})/\sigma_p$, which contains no λ either; only a balanced mixture ($\lambda\sigma_s \sim \sigma_p$) admits a λ trend.

Proof. Substitute $R_i = C + \lambda s_i - p_i$ into Eq. 1 and take the respective limits. □

Proposition 1 says the z-score *cancels the shaping coefficient*: an ε-scale signal is stretched to full z-scale advantages per pooled step-sample (the measured $\pm$5–9 fingerprint, within the $\approx$14 bound above). *Within the $\lambda\sigma_s \gg \max(\sigma_p, \epsilon)$ regime*, no reduction of λ changes the update; once $\lambda\sigma_s \sim \max(\sigma_p, \epsilon)$ the shaping term no longer controls the pool and the first limit gives way, so the statement is conditional, not absolute. With $\epsilon = 10^{-6}$ (the framework default), the regime holds while the signal retains within-group variance; at saturation $\sigma_s \to 0$ and the shaping contribution vanishes (Proposition 2). This one line predicts two otherwise separate facts: the anneal arm collapsing unchanged despite λ having already fallen to $\approx$0.068 by the turn (§5.2), and the registered prediction that std runs behave identically across $\lambda \in \{0.01, 0.03, 0.1\}$. The sweep confirms all three points:

amplified scales $\pm 7.3/\pm 9.1/\pm 7.2$ (no trend over a tenfold range), collapse in every arm (endpoints 1.0/0.0/1.0%), timing within the variance band. The coefficient is cancelled; what remains is the variance structure.

Proposition 2 (Saturation-endpoint case of the variance criterion)**.** *In the setting of Proposition 1 (an all-fail group with $R_i = C + \lambda s_i - p_i$; the dominance premise is not inherited), if the signal saturates so that its within-group deviations vanish on the scale of the remaining spread ($\lambda\sigma_s \ll \max(\sigma_p, \epsilon)$), the* shaping component's *contribution to the advantage vanishes,*

$$\frac{\lambda\,(s_i - \bar{s})}{\sigma_{\lambda s - p} + \epsilon} \;\longrightarrow\; 0, \tag{4}$$

and the amplifier can no longer feed on the signal. If the penalty spread also vanishes ($\sigma_p \to 0$), the ϵ-floor reactivates and the amplifier starves outright; otherwise the z-score is taken over by the penalty term (the second limit of Proposition 1), amplified pressure that rewards surface compliance rather than the signal, the same pressure every healthy baseline carries. In either case the danger a dense signal itself contributes under std normalization is governed not by its magnitude or density but by its within-group variance at saturation*.*

3.3 THE VARIANCE-TRAJECTORY CRITERION

What z-scoring amplifies is the signal's within-group *variance*, so we state the danger condition over the whole variance trajectory, not only its endpoint: **a dense signal is dangerous under std normalization to the extent that it sustains within-group variance during the phase when that variance dominates the group advantage**. All-fail domination is the sparse-success route into that phase (our collapse configuration); the WebShop easy regime shows a second route, where continuous task scores over short episodes let prediction variance dominate from step 0 (§6). The window is not defined by the harm it produces, but our ex-ante observability is uneven and we state it exactly. The all-fail group fraction was measured live in the two group-size ablations (its climb precedes their literal-zero windows) and retrodicted for the original run (§5); the shaping term's share of the pooled step-sample spread is implementable as a monitor (the terms enter the reward separately) but our runs logged only batch means (§3.2); and for the continuous-score route neither proxy applies, so the warning for that regime (§6.5) rests on the measured failure itself, not on a monitored precursor. For saturating signals the endpoint (σ_{within} at mastery) is the limiting summary, but the operative quantity is the variance carried through the variance-dominated window; a progress-style signal has its *largest* variance while the agent is still learning, so the criterion does not declare it safe during training, only that its pressure decays as mastery approaches. Empirically the signal forms separate as the criterion predicts: difference-form variance persists to the absorbing state (collapse); saturated always-positive confidence starves the signal-fed amplification (no collapse); progress-style Δacc plateaus below saturation and keeps paying (chronic drag: the endpoint form of the criterion would have called it safe, the trajectory form does not); anchor-QA over-warns (drag at training caliber only). A matched-variance noise control adds the second axis: sustained variance plus *hackability* is necessary in every reward-channel lock we observe but not jointly sufficient. The full walk-through is in Appendix N. The criterion is consistent with published reward-channel successes in the weak sense that neither contradicts it (RWML is level-scored, outside the difference-form family; VAGEN's estimator coordinates are unreported, and our GiGPO arm shows redesign alone does not decide); it issues no safety certificates, and the full compatibility discussion is in Appendix N. Proposition 2 formalizes the criterion's *endpoint* case; the trajectory form, which tracks σ_s through the variance-dominated window rather than only at saturation, is its empirical strengthening, the one the Δacc and anchor-QA arms forced. The saturation quantity is the same one that lower-bounds the policy-gradient magnitude in Leng et al. (2025). This is a signal-side taxonomy, orthogonal to algorithm-side variance work that redesigns the normalizer or its weights (Xiao et al., 2025; Tao et al., 2025); its tests were preregistered before the runs completed.

4 EXPERIMENTAL SETTING

Environments and framework. Three environments carry results, all driven by the step-independent multi-turn agent stack of §3.1 on the open-source verl-agent framework (hardware and per-environment configuration in Appendix C). ALFWorld (Shridhar et al., 2021) is the primary testbed (sparse terminal success 10 / failure 0; horizon 50); WebShop (Yao et al., 2022) replicates the matrix core under a *continuous* terminal matching score in two regimes (small: 1k products, baseline $\sim$60%; full: 1.18M products with human goals, baseline $\sim$29%; horizon 15); HiddenRule-Gym is our synthetic rooms-and-devices POMDP, described below.

Models and configuration. Qwen3-1.7B/4B/8B, identical configuration across scales (train batch $8 \times$ group $4 = 32$ trajectories/step; validation 32 episodes, 16 for HiddenRule-Gym at 1.7B with the 4B HRG capacity arm at 32, sampled decoding at temperature 0.4). Endpoints are last-6 validation means (single-point binomial noise up to $\pm$8.8pt; last-6 SE $\pm$3.4pt from the measured single-point σ of 8.4), seed 0 unless stated (the headline auxiliary-loss arm, both leading placebos, and the ALFWorld baseline carry seeds 0/42/96); formal endpoint numbers from a unified 140-game evaluation of every arm entering a cross-arm ordering claim are reported in §6.4 (Table 4), and prose keeps last-6 numbers where it describes training dynamics.

Budget. All arms run at one quarter of published full-budget ALFWorld training (32 trajectories/step, 150 steps) so fourteen mutually comparable arms fit one compute envelope; every claim is a same-budget arm-to-arm comparison and none depends on absolute success rates (published baselines, a full-budget sanity anchor at 67.2%, and state-of-the-art reference points are in Appendices H and D).

HiddenRule-Gym (HRG). A synthetic rooms-and-devices POMDP with four hidden-rule families (conjunction, sequence, XOR, count; train/probe family split), a BFS oracle sharing the environment's pure transition function, and (uniquely) *exactly computable* feature coverage $C = I(\Phi; s)/H(s)$ over non-terminal reachable states, with a greedy mask ladder giving measured coverage levels for the sweep in §6.3. Privileged-latent belief probes carry a 5-gram leakage audit.

Arms, baselines, and evaluation calibers. Every claim is a same-budget arm-to-arm comparison against a matched GRPO baseline trained under the identical configuration (three seeds at 4B on ALFWorld; per-scale and per-environment baselines elsewhere); the study comprises 74 training arms, inventoried with endpoints in Appendix F, and its predictions were preregistered by SHA256 digest in a public version-controlled file before the corresponding runs completed (ledger in Appendix A). Two evaluation calibers recur throughout: *training caliber* (the last-6 validation means above, used for training dynamics) and *formal endpoints* (the unified 140-game re-evaluation of §6.4, used for every cross-arm ordering claim; WebShop uses the analogous 200-goal protocol).

5 Experimental Results

5.1 Three-scale collapse and the mechanism, measured

Sparse reward $\rightarrow$ all-fail groups $\rightarrow$ the only within-group differences are ε-scale $\lambda\, r_{\text{pred}}$ $\rightarrow$ std normalization amplifies them to full scale $\rightarrow$ the gradient rewards *predictability* $\rightarrow$ the policy drifts toward familiar states $\rightarrow$ fewer successes $\rightarrow$ stronger domination: a self-reinforcing loop, and at saturation $r_{\text{pred}} \rightarrow 0$ strands the policy in the degenerate basin. The advantage fingerprint makes the amplification visible: all-fail-group advantages reach **±5–9** under std normalization (inside the pooled step-sample bound of $\approx$14, §3.2) versus the unamplified ± 0.1 scale ($= \lambda$) without it. Replay forensics on a collapsed checkpoint, constant-budget group-size ablations ($n \in \{4, 8, 16\}$: literal zero at steps 85/135/80, non-monotone), and the all-fail-fraction telemetry (a climb to a terminal lock at 1.00 coinciding with the literal-zero window) confirm each link of the loop, and a tenfold λ reduction leaves the amplified scale

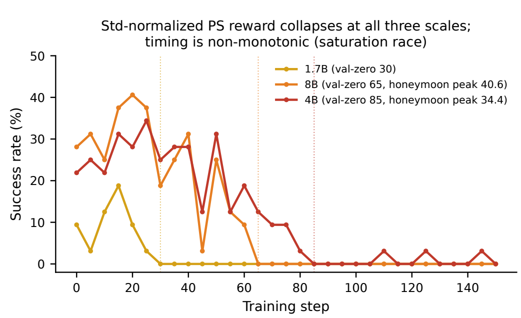

Figure 2: The std-normalized prediction reward collapses every run at all three scales; timing is non-monotonic (saturation race, §5.2). Honeymoon peaks *rise* with scale: the signal lifts each arm's own early trajectory until hacking pressure arrives. Neither honeymoon peak exceeds its arm's formal baseline (8B $40.6 < 45.0$; 4B $34.4 < 42.9$); the 1.7B curve's single transient point (18.8, not a honeymoon by the Table 1 criterion) sits at its formal baseline's level (17.9).

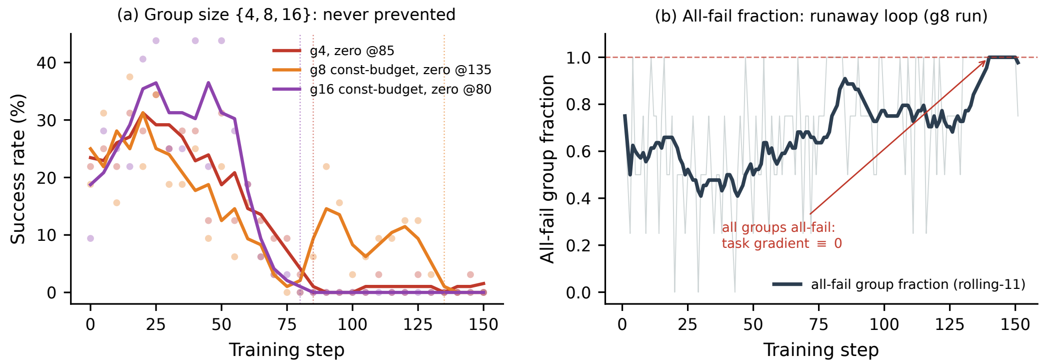

Figure 3: Left: constant-budget group-size ablations. Collapse occurs at every group size in $\{4, 8, 16\}$ with non-monotone timing (literal zero at steps 85/135/80): group size neither prevents nor reliably delays. Right: the all-fail group fraction along the n=8 run (rolling mean over 11 steps); its terminal lock at 1.00 coincides with the literal-zero validation window: the absorbing state of the self-reinforcing loop measured directly (the n=16 run shows the same lock from step $\sim$75).

unchanged ($\pm$7.2 vs. $\pm$7.3), as Proposition 1 predicts; Figure 3 shows the group-size ablations and the all-fail lock, and full measurement detail is in Appendix H.

scale	honeymoon peak	turn	val-zero from	terminal triple
1.7B	—	$\sim$15–20	30	pred 1.000 / len 50 / succ 0
4B	34.4 @25	$\sim$45	85	same
8B	**40.6 @20**	$\sim$48	65	same

Table 1: Collapse phenomenology. *Turn* = behavioral inflection (validation begins its monotone decline); *val-zero* = first checkpoint at literal 0% success. These are distinct quantities, and the saturation-race ordering in §5.2 refers to val-zero. Baselines for reference (last-6, log-exact): 1.7B 20.9%, 4B 49.5%, 8B 32.8%. *Honeymoon* is operationalized as at least two consecutive validation points above the matched baseline's contemporaneous curve; the 1.7B arm's single transient point (18.8% at step 15, inside the $\pm$8.8pt single-point noise) does not qualify, hence "absent" there.

5.2 Saturation race, irreversibility, and early warning

A saturation race, not a coefficient response. We registered "weaker model collapses earlier, 8B latest" and the data falsified it: 1.7B@30 < **8B@65** < 4B@85 (val-zero caliber throughout; Table 1). Revision: collapse timing is a *race* between prediction-saturation speed and task-learning speed; the 8B model learns everything faster, including how to be predictable. The honeymoon peak rising with scale (34.4 $\rightarrow$ 40.6) shows the early transient scales with capacity; note it lifts the arm's own trajectory only, and neither honeymoon peak exceeds its arm's formal baseline (the 8B peak tops the artifact-depressed training-time 32.8, not the formal 45.0; the 1.7B single transient, no honeymoon by the Table 1 criterion, sits at its formal baseline's level, 18.8 vs. 17.9).

Irreversibility. Nothing recovers a collapsed policy: 120 post-collapse steps at fixed λ (1.7B), cosine annealing to zero (the turn arrives at λ≈0.068, step $\sim$57; the final $\sim$30 effectively-pure-GRPO steps show zero recovery), and a preregistered post-hoc normalizer switch (resuming the seed-42 arm 70 steps inside its absorbing state: 11 checkpoints at literal 0%, the amplifier verifiably gone, magnitudes $\pm$5–9 $\rightarrow$ ≈0.1) all fail; post-collapse entropy *rebounds* at all scales (0.39/0.44/0.52 ordered by reference-model entropy), so the absorbing state is behavioral, not temperature-level. Three preventive routes work: mean-only from the start, all-fail-group filtering (partial, 39.6%), and

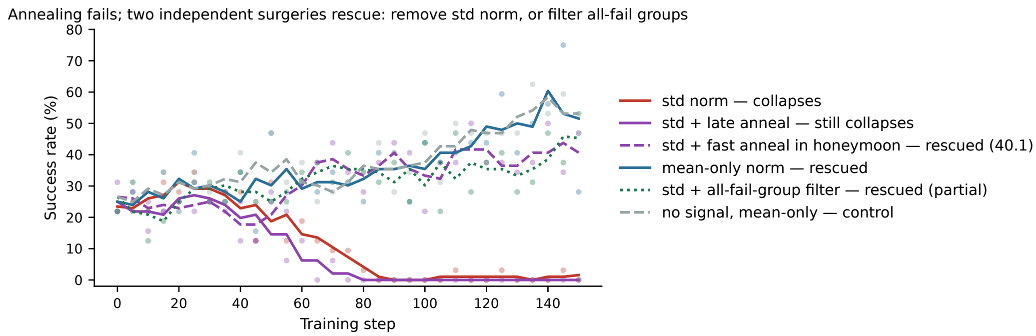

Figure 4: Two independent interventions rescue. The collapsed arm and the mean-only arm differ only in `norm_adv_by_std_in_grpo`; the filter arm (dotted) keeps std normalization and instead drops all-fail groups from the gradient. At step 85, the unfiltered arm's literal-zero step, it sits at 31.2%, and it finishes at 39.6% (last-6). Late annealing does not rescue, but the same anneal applied *inside* the honeymoon window does (dashed, 40.1%): withdrawal timing relative to the honeymoon decides. The no-signal control matches the mean-only arm (net attribution ≈ 0).

timed withdrawal (a fast-anneal arm completing inside the honeymoon window survives at 40.1%, Fig. 4). Irreversibility binds after the turn, not before; full detail in Appendix H.

Early-warning signature. A frozen conjunction rule (entropy-has-declined $\wedge$ prediction ≥ 0.95 $\wedge$ length $\geq 0.95\,H$; thresholds calibrated only on the four definition-set collapse runs) detects all three held-out *reward-channel* collapses **73/53/13 steps ahead** of the collapse point, with at most one false alarm among the eleven training-healthy arms; the full rule, roster, three-caliber definitions, and independence caveats are in Appendix H.

5.3 Single-factor rescue and the fix hierarchy

The rescue. Flipping one flag (advantages centered by group mean but not divided by group std) turns 0% into **51.6%** ($\geq$ baseline 49.5%), peak 75.0, no terminal triple, prediction accuracy does not saturate (0.49–0.64). This is the causal localization: the std term is necessary for the catastrophe.

The second intervention. A preregistered companion arm keeps std normalization untouched and instead drops all-fail groups from the gradient (RAFT-style filtering; 23–77% of samples zeroed per step). It also prevents collapse (no terminal triple; prediction accuracy oscillates at 0.6–0.8) and finishes at **39.6%**: rescued from 0% but $\sim$10 points below baseline, consistent with zeroing a large, fluctuating share of the samples (23–77% per step) and with residual pressure inside mixed groups. The two interventions complete an ALFWorld knock-out matrix: remove the signal (baseline), the amplifier (mean-only), or the all-fail groups (filter) and training is healthy; only the full triple collapses. (On WebShop, where the terminal score is continuous and strict all-fail groups do not form in either mode, the all-fail leg does not apply; prediction variance dominates through low return spread instead, §3.3.) The asymmetry (mean-only restores parity, filtering rescues partially) makes mean-only the preferred prescription, the filter arm independently confirming that the pathology lives in the amplification of all-fail groups, the same groups DAPO-style dynamic sampling discards for variance reasons (Yu et al., 2025).

Net attribution. The control (mean-only normalization, *no* prediction signal) lands at **52.6%**, dead even with 51.6% (-1.0pt, far inside single-seed validation noise; descriptive comparison): the prediction reward's net contribution in the mean-normalized channel is ≈ 0, and the rescue belongs to the normalizer. Secondary finding: mean-only $\geq$ std baseline independently replicates Dr.GRPO in long-horizon agents, with a caliber-dependent margin ($+3.1$ training on the matched-seed pair, 52.6 vs. 49.5; formally $+13.8$ against the three-seed mean and $+15.0$ on the matched seed-0 pairing, 57.9 vs. 44.1/42.9). Residual: the PS arm sharpens $\sim$20 steps earlier at equal endpoint ("speed, not height"; single seed).

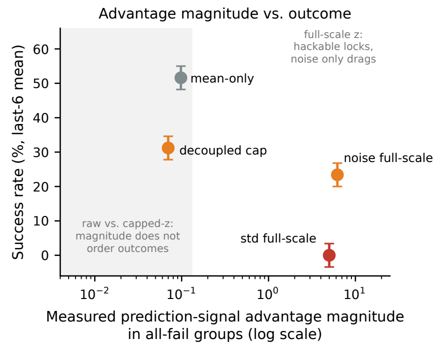

Figure 5: Advantage magnitude versus outcome in all-fail groups (log x). Magnitude does not order the outcomes: the harmless mean-only arm's raw contribution (± 0.098) exceeds the dragging decoupled arm's capped one (≈ 0.07). The split tracks amplification structure (raw / capped z-scored / full-scale z-scored) and, at full scale, hackability (matched-variance noise drags but does not lock).

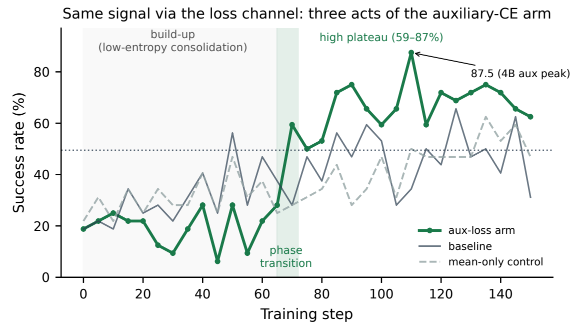

Figure 6: The auxiliary-loss arm's three phases: low-entropy build-up (entropy 0.04, a false "erosion" alarm), phase transition (steps 65–72, 28$\rightarrow$59%), high plateau (59–87%; 4B aux-arm peak 87.5). The interference probe (task-batch log-prob shift across the auxiliary update) stays at zero throughout: the auxiliary gradient coexists cleanly with the task gradient.

Two further arms sharpen the criterion and close the fix list. A preregistered matched-variance *noise* control (random potentials, same per-step variance, not policy-controllable) drags severely (23.4%, -26 vs. baseline) yet never locks: hackability is the criterion's second axis, and the two axes are necessary in every observed reward-channel lock but not jointly sufficient (the Δacc and anchor-QA arms satisfy both and neither locks). Per-channel *decoupling* (GDPO-style, contribution capped at λ) does not collapse but under-performs (31.2/42.2 last-6; formal 27.9/32.1, both below matched baselines): the per-channel z-score is a capped copy of the same amplifier, and advantage *magnitude* does not order the outcomes, amplification structure does (the harmless mean-only arm's raw ± 0.098 exceeds the dragging capped arm's 0.066–0.077). Fig. 5 plots magnitude against outcome; full paragraphs are in Appendix I.

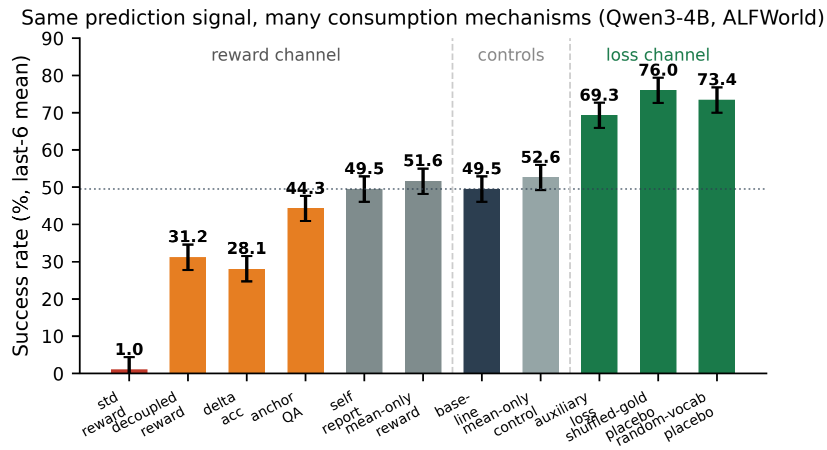

Figure 7: The signal-delivery matrix (11 of the 14 arms of Table 2 shown; the all-fail filter, compute-matched replay, and format-stripped placebo omitted for width): same prediction signal, varying consumption mechanism and normalization, identical environment/model/budget. Regime membership tracks the delivery channel rather than the signal form. Numbers here are training-time last-6 means (the quantity the dynamics narrative tracks); formal 140-game endpoints for every arm are in Table 4.

5.4 The channel effect

arm	channel	signal form	normalization	last-6
std reward	reward	diff.-form	std	1.0%
decoupled reward	reward	diff.-form, per-channel cap	std (decoupled)	31.2%
Δacc progress	reward	clipped accuracy progress	std	28.1%
mean-only reward	reward	diff.-form	mean-only	51.6%
all-fail filter	reward	diff.-form	std, all-fail groups dropped	39.6%
mean-only control	—	*no signal*	mean-only	52.6%
self-report	reward	always-positive confidence	std	49.5%[1]
anchor-QA	reward	recall accuracy	std	44.3%
baseline	—	—	std	49.5%
compute-matched replay	—	no aux target, $2\times$ epochs	std	48.9%
auxiliary loss	**loss**	teacher-forced CE on gold predict	std untouched	**69.3%**
shuffled-gold placebo	**loss**	CE on permuted gold	std untouched	**76.0%**
random-vocabulary placebo	**loss**	CE on random out-of-domain strings	std untouched	**73.4%**
random-token placebo	**loss**	CE on bare random words (format-stripped)	std untouched	**61.5%**

Table 2: The signal-delivery matrix. On class means the loss-channel arms beat every reward-channel variant with the std reward channel left untouched; the shuffled-gold placebo matching the true-gold arm shows exact content–context pairing is not the active ingredient (see text). The auxiliary-loss arm carries three seeds (69.3/63.5/50.0, mean 60.9), and both leading placebos now carry three seeds of their own (shuffled 76.0/60.4/48.9; random-vocabulary 73.4/77.6/60.9, training-caliber): five of the six placebo seeds land above the baseline band and the sixth inside it (none below, none collapsing; class means 66.2 vs. 60.9), so the class-level ordering (placebo $\geq$ gold) is seed-replicated. Remaining single-seed cells are marked in the text.

Four confounds separate the auxiliary-loss arm from baseline (prompt format, predict-block generation, extra update, signal content). The mean-only pair differs in signal presence, not prompt alone, so it bounds the *joint* effect of the first two plus the mean-channel signal at $\approx$0 at 4B, assuming no mutual cancellation (the study's only direct prompt-only arm, at 8B, shows a $+7.1$ point-estimate gain with overlapping intervals, Table 4, so the prompt-side term is not separately measured and the

[1]The self-report arm's last-6 mean coincides with the baseline's (both 49.5%) by arithmetic accident: the two arms' final six validation points differ individually but share the same sum. Verified against raw logs; not a transcription error.

bound is channel- and scale-local); the shuffled-gold placebo preserves compute/update-count/mask structure while destroying content–context pairing, separating the third from the fourth. The placebo verdict landed on the informative side: shuffled-gold *matches* true gold (76.0 vs. 69.3; descriptively indistinguishable), so exact content–context pairing is *not* the active ingredient, and the channel gap survives strengthened: on class means the loss-channel arms beat every reward-channel variant *without requiring correct labels* (the weakest gold seed, 50.0, lands level with the mean-only arms' 51.6–52.6). Seed replication of the true-gold arm (69.3/63.5/50.0 at seeds 0/42/96) sets the effect size, $+\mathbf{14.2}$ on matched-seed pairing (SE 8.0; $+19.8/+24.4/-1.6$), every seed above the three-seed baseline mean though the third sits 1.6 below its matched baseline, with three qualitatively different phase-transition timings (steps $\sim$70, $\sim$100, $\sim$25) all converging to an elevated plateau.

Two further controls complete a graded attribution ladder (full detail in Appendix J): the preregistered random-vocabulary strong placebo reaches **73.4%**, statistically tied with gold (69.3) and shuffled gold (76.0); a compute-matched replay control gains nothing (48.9); and a format-stripped random-token placebo captures a majority of the effect (61.5). The ladder reads baseline 49.5 $\approx$ compute replay 48.9 $<$ random-token 61.5 $<$ schema placebos 73.4–76.0 $\approx$ gold 69.3; compute contributes nothing and content contributes nothing *positive* at any level (formally, the gold class mean sits below the placebo class mean at both matched seeds, §6.4). Figure 6 shows the headline arm's three-phase dynamics and its zero-interference probe. A preregistered gradient-noise falsifier (structure-matched Rademacher $\pm\beta$ sign noise) lands at 43.2% last-6, inside the baseline band: the gain requires *directed* novel-target gradients, not update noise. Stated at its strongest: on class means, a format-consistent auxiliary CE pass on *arbitrary* strings outperforms every reward-channel delivery of a genuinely informative signal; gains attributed to world modeling in prior work may be substantially attributable to the auxiliary update itself.

6 Discussion: Boundaries, Ablations, and Formal Evaluation

Two extensions probe the boundary of the channel result: the same auxiliary-loss recipe at 8B, and the matrix core replicated on a second environment.

6.1 At 8B the loss channel turns bistable

Running the auxiliary-loss recipe unchanged on Qwen3-8B (same 8×4 configuration, 150 steps) breaks the no-seed-below-baseline pattern (a formal-caliber pattern at 4B: 68.6/57.9/52.1 all above their matched 42.9/40.7/48.6; at training caliber the third 4B seed already sat 1.6 below its matched baseline), in both directions. Seed 0: a three-point early transient (28.1/34.4/25.0, comparable to the baseline's contemporaneous band rather than above it, so not a honeymoon in the Table 1 sense) collapses at steps 15–20 into *27 consecutive zero-success evaluations* (130 steps; the deepest absorbing state we measure), with the full terminal phenotype: entropy $0.45\rightarrow0.03$, per-step *response* length $179\rightarrow36$ tokens (episode length does not pin here: this failure class bypasses the length-pinning channel, consistent with the criterion not covering it), prediction-parse validity and all-fail-group fraction both pinned at 1.000. Seed 42: the *same* collapse onset (three zero evaluations, steps 20–30), then spontaneous escape and a climb to a last-6 mean of **72.4%** (peak 90.6; formal **85.0%**, $+\mathbf{40.0}$ over the formal 45.0% baseline, the study's best gold-signal endpoint). Same recipe, data, and budget; 72 points of outcome variance across two seeds. This is not generic fine-tuning seed variance (Dodge et al., 2020; Mosbach et al., 2021): two of the three seeds enter the same early collapse window (steps 15–30) and the third locks later ($\sim$step 85, from the lead), yet every locked outcome shows the same absorbing phenotype, so the outcomes separate into two mechanistically distinct attractors reached by different routes rather than a diffuse spread (cf. scale-emergent instabilities in pretraining, Wortsman et al., 2023). Two observations matter for the paper's claims. First, this collapse occurs *without* the reward channel or z-score amplification (auxiliary CE pressure under task-gradient starvation), a failure class the variance-trajectory criterion does not cover. Second, the loss channel's no-harm record is scale-qualified: at 4B the auxiliary channel was safe in ten of ten completed ALFWorld runs; at 8B gold full-weight locks two of three seeds and sends the third to the study's best gold-signal result, a scale dual of the reward channel's 4B law (there: collapse certain, timing random; here: occurrence itself random).

Family controls localize the lock (Fig. 8; full detail in Appendix K): shuffled-gold climbs healthy to 78.6 (formal 79.3, level with the escape seed's 85.0), so the *gain* side of the content-free attribution

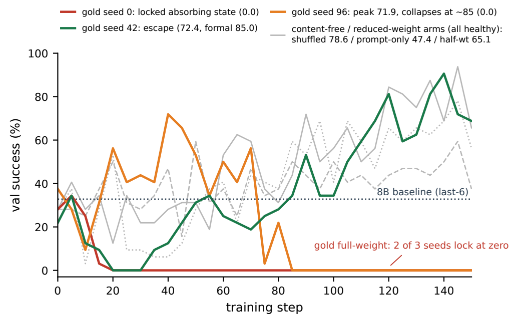

Figure 8: The 8B family. Gold full-weight auxiliary CE is bimodal across seeds: two of three seeds lock at literal zero (one after leading the field at 71.9%), one escapes to the study's best gold-signal endpoint (85.0%). Every content-free or reduced-weight arm (shuffled, prompt-only, half-weight β=0.05; gray) stays healthy. The risky ingredient at 8B is the true-gold signal at full weight, not the auxiliary pass itself.

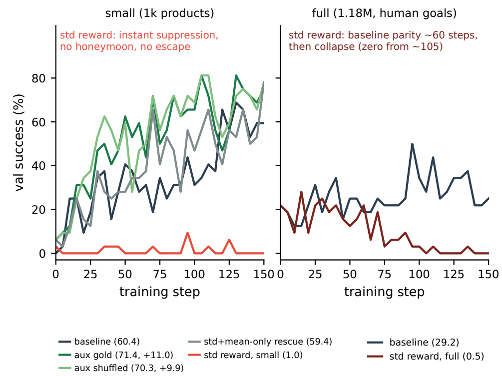

Figure 9: WebShop validation curves. Left: small mode, where the std reward arm is suppressed from step 0 (no honeymoon) while the auxiliary-loss arm beats the baseline by +11.0 training caliber (formal +3.0, statistically level). Right: full mode, where the std arm holds baseline parity for ~60 steps, then collapses to a sustained floor, reproducing the ALFWorld collapse arc (turn, then sustained zero) with the honeymoon replaced by baseline parity, on a comparable timescale.

carries to 8B; prompt-only stays healthy (47.4), so the prompt alone explains nothing; a half-weight arm on the locked seed escapes (65.1), so the coefficient participates causally in basin entry (no tension with Proposition 1: β never passes through advantage normalization); and the third gold seed locks from the lead (71.9 $\to$ 0 at $\sim$85). Gold full-weight locks two of three seeds; every content-free or reduced-weight arm stays healthy; under a content-blind null the two healthy auxiliary-CE arms have probability $(1/3)^2 \approx 0.11$ (0.04 counting prompt-only), suggestive but six samples remain six (§6.6).

6.2 On WebShop the reward channel fails again; the loss-channel gain does not confirm

arm (WebShop, 4B, last-6)	small (1k products)	full (1.18M, human goals)
baseline	60.4%	29.2%
std prediction reward	**1.0%**	**0.5%**
auxiliary loss (true gold)	**71.4%** (+11.0)	—
auxiliary loss (shuffled gold)	**70.3%** (+9.9)	—
std reward + mean-only rescue	**59.4%**	—

Table 3: The matrix core on a second environment (Figure 9). Both std reward arms fail, in mode-dependent forms: the small-mode arm does not leave the floor (max 9.4%, no honeymoon, no escape, valid-action ratio 1.000, so not a parsing artifact); the full-mode arm tracks its baseline for $\sim$60 steps, then collapses (sustained zero from step $\sim$105). The auxiliary-loss arm's training-caliber gain (+11.0) compresses to +3.0 under the formal 200-goal evaluation (§6.4 protocol family), statistically level with its baseline. Success counts episodes the environment marks `won` (a perfect terminal matching score); the training reward is the continuous score itself (Appendix C), and the 200-goal formal protocol uses the same success definition.

The reward-channel failure generalizes with regime-dependent form (a four-form taxonomy: honeymoon-collapse on ALFWorld; collapse-after-parity on full-mode WebShop, within the 30–135 val-zero range the ALFWorld arms span with group-size ablations included; instant suppression on small-mode WebShop; chronic drag on the capped arms), the attribution and the rescue both transfer (shuffled placebo 70.3 vs. gold 71.4, formally 67.0 vs. 63.0; mean-only 59.4, a $+\mathbf{58}$-point rescue over the std arm, a cross-normalization comparison that could only flatter the arm and it still does not beat the baseline), and the auxiliary gain is seed-split ($+11.0/-9.9$ training; $+3.0/-14.0$ formal), a single-seed-pair difference, not a confirmed effect size. On HiddenRule-Gym the same auxiliary recipe is statistically inseparable from the no-signal arm (17.1 vs. 10.7) and below the coverage-matched reward arm: the loss-channel gain has conditions that one environment success plus a seed-split second do not license extrapolating away. Estimator controls close the localization: GiGPO (step-level grouping, group-std kept) collapses on schedule; PPO/GAE (37.5%), RLOO (40.6 vs. its own 44.8 baseline), and REINFORCE++ (global-batch whitening, completed after the main freeze: 48.4 vs. its own 55.2 baseline, no collapse window) survive the identical recipe; collapse tracks whether the advantage *divides by a within-group* σ, with the signal necessary (§5.3) but not by itself deciding. Full curves, per-arm paragraphs, and the estimator qualifiers (all training-caliber, single-seed; the REINFORCE++ signal arm post-dates the freeze) are in Appendix K.

6.3 Separating coverage, dynamics, and capacity

HRG's exactly computable coverage separates what ALFWorld conflates: the coverage axis (no-signal floor with gradient starvation; non-covering Φ erodes belief probes; the coverage sweep pays at mid coverage, Fig. 10) and the capacity axis (4B lifts off the 1.7B floor without cracking rules; training caliber). Full arm-by-arm detail is in Appendix K.

Failure matrix. Low coverage = wrong beliefs (B/C); adequate coverage + bad dynamics = collapse (ALFWorld + std); coverage + clean dynamics + insufficient capacity = partial conversion far below mastery (arm D at 1.7B: 24.3% vs. 10.7% no-signal at the 140-game endpoint, a gap partly attributable to the cross-normalization difference noted above); all three satisfied = parity (rescued arm at 4B ALFWorld). The gain regime requires switching channels (§5.4). Figure 10 plots the coverage sweep at its formal endpoints.

6.4 Formal endpoint evaluation

Last-6 training-caliber means (32-game sampled validation, binomial noise $\approx \pm$ 8.8pt; 16-game for HiddenRule-Gym, $\approx\pm$12.5pt) are adequate for shape and large qualitative gaps but not for cross-arm ordering. We therefore re-evaluate the final checkpoint of every arm entering a cross-arm ordering claim (the arms of Table 4, all 11 HiddenRule-Gym arms, and the WebShop arms; the mechanism battery keeps training-caliber endpoints, §6.6, and its arm-to-arm statements are deliberately not treated as formal orderings) under a single protocol: the full 140-game `valid_seen` split and the 134-game `valid_unseen` split (padded to 140 episodes by the fixed-batch loader, six games run twice), same decoding (sampled, temperature 0.4), same hardware, one pass per arm (Table 4). All 11 HiddenRule-Gym arms get the same treatment (140 games on the held-out probe rule families, fixed seed); corrections there reach +13.9 points, twice revising a verdict (arm D, §6.3; the full-coverage rung, Fig. 10), and Figure 10 plots the sweep with its formal intervals. WebShop arms use the analogous 200-goal protocol (test goals 0–199, one episode each; formal numbers appear in the text and Appendix F, training-caliber numbers in Table 3).

Four things survive formalization, one is revised, and one weakens. (i) The channel dichotomy sharpens. From the 44.1% baseline three-seed mean, the compute-matched control sits at baseline (48.6, a +4.5 point-estimate difference inside the interval). The auxiliary gain against its *matched-normalization* baseline is $+\mathbf{15.4}$ (gold three-seed formal mean 59.5; on the matched seed-0 pairing the gap is categorical by the caption's own bar: 68.6 vs. 42.9, $+\mathbf{25.7}$, intervals disjoint), and the content attribution runs entirely within that matched machinery: under the identical std-plus-auxiliary recipe, the content-free placebos sit at or above the gold mean (67.9–87.1 vs. 59.5; the seed-paired form of this ordering is given in (ii) below). A separate, cross-normalization observation: fixing the normalizer alone, with no signal of any kind, reaches the same level (mean-only no-signal, 57.9), so the auxiliary pass is not the only road to $\sim$58; we flag this convergence as descriptive, not a matched contrast (the arms differ in normalizer, exactly the comparison this section otherwise disallows), and the combined cell (mean-only *plus* auxiliary) is untested. The separation runs along the delivery machinery; the supervision content contributes nothing positive. Meanwhile no reward-channel variant sits measurably above its matched-normalization baseline; several point estimates sit below

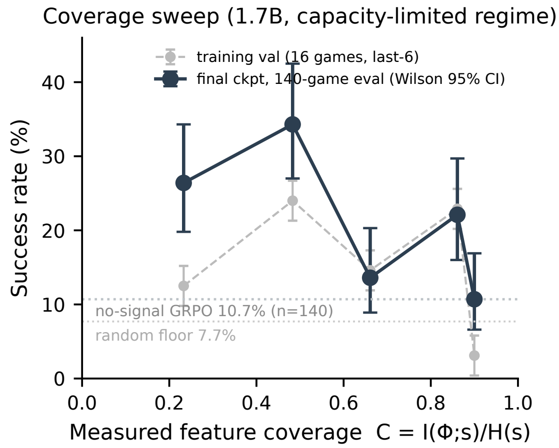

Figure 10: Coverage sweep at measured C (greedy mask ladder), all five rungs at the 140-game endpoint protocol (Wilson 95% CIs): 26.4 / **34.3** / 13.6 / 22.1 / 10.7%, against the no-signal GRPO arm's 10.7% on the same 140 games (dotted line; the gray dashed curve is the training-time 16-game series, whose state-sampling error reached $+13.9$ points). The endpoints tell a two-regime story. Left: coverage buys success; the peak rung's interval [27.0, 42.5] is fully separated from the no-signal reference (a 23.6-point gap: interval-separated yet under the $\geq$25-point categorical bar of §6.4, and cross-normalization per the caveat below). Right extreme: at full coverage the prediction target itself overwhelms 1.7B capacity: prediction accuracy plateaus at 0.70–0.80 far below ceiling, terminal entropy drifts to the random-policy level (0.44), and success returns to the no-signal level (10.7%), the coverage benefit fully consumed, under mean-only normalization, i.e., independently of the std amplification pathology. The middle rung's dip (13.6) overlaps its neighbors' intervals and we leave the mid-coverage shape unresolved. All rungs single-seed. One design caveat: the no-signal reference is std-normalized while the coverage arms are mean-only, so the gaps fold in a normalization difference. On ALFWorld the same normalizer change measures $+3.1$ at training caliber but $+13.8$ at the formal endpoint (§5.3), so this component cannot be assumed small, and the interval separations above license only arm-versus-arm differences, not their attribution to coverage alone; a mean-only no-signal arm would separate this cleanly and is left to future work.

it (self-report -5.5, Δacc -12.0, decoupled -16.2 against the 44.1% mean), though these gaps stay under the caption's categorical threshold with intervals overlapping the baseline's. The collapsed arms are confirmed at or statistically indistinguishable from literal zero (upper CI bounds $\leq$3.9%). Pairings are stated per comparison throughout this section: gaps in this paragraph use the three-seed baseline mean (44.1), the seed-0 ladder in (ii) uses the matched seed-0 baseline (42.9). (ii) The content-free attribution is the study's most consistent formal pattern, stated within scale: at 4B the shuffled-gold s0 placebo (81.4) and random-vocabulary arm (87.1) sit above every true-gold seed (68.6/57.9/52.1), while the shuffled s42 and random-token endpoints (67.9) do not; the surviving statement is a descriptive, seed-paired class ordering, not a categorical claim (full ladder restatement and calibration notes in Appendix L). (iii) The mean-only rescue and its no-signal control stay within overlapping intervals (52.9 vs. 57.9, a 5.0-point gap at point estimate, the same evidentiary grade as the self-report gap), preserving the net-zero attribution of the cleaned reward channel. (iv) The compute control stays at baseline (48.6; $+4.5$ at point estimate, inside the interval, the same evidentiary grade as the negative point-estimate gaps above). The revision: the 8B baseline evaluates at **45.0%** [37.0, 53.3], inside the 4B baseline seed band, so the training-time anomaly (32.8% $<$ 4B) was substantially a 32-game-window artifact, and scale-ordering caveats tied to it are lifted. Family-arm gains restate on formal numbers against the 45.0 baseline (gold escape seed: $+40.0$; shuffled: $+34.3$; both intervals disjoint from the baseline's).

Out of distribution and the largest revision. On the 134-game unseen split every headline structure survives and every magnitude compresses (collapsed arms literal zero, upper bounds 2.7%; auxiliary gain $+15.4 \rightarrow +7.9$; four near-baseline gaps flip sign within noise); the largest downward revision is self-report ($49.5 \rightarrow 38.6$), the reason single-seed rows carry a cross-seed caveat. Full out-of-distribution and calibration prose is in Appendix L.

arm	channel / role	seen (140)	[95% CI]	unseen (134→140)	last-6
1.7B					
baseline	—	17.9	[12.4, 25.1]	18.6	20.9
std prediction reward	reward	0.0	[0.0, 2.7]	0.0	0.0
4B					
baseline (s0)	—	42.9	[35.0, 51.2]	42.1	49.5
baseline (s42)	—	40.7	[32.9, 49.0]	37.1	39.1
baseline (s96)	—	48.6	[40.5, 56.8]	46.4	51.6
aux loss, gold (s0)	loss	68.6	[60.5, 75.7]	65.0	69.3
aux loss, gold (s42)	loss	57.9	[49.6, 65.8]	45.0	63.5
aux loss, gold (s96)	loss	52.1	[43.9, 60.2]	39.3	50.0
aux loss, shuffled gold (s0)	loss placebo	**81.4**	[74.1, 87.0]	**73.6**	76.0
aux loss, shuffled (s42)	loss placebo	67.9	[59.8, 75.1]	66.4	60.4
aux loss, random vocab (s0)	loss placebo	**87.1**	[80.5, 91.7]	71.4	73.4
aux loss, random tokens (s0)	loss placebo	67.9	[59.8, 75.1]	57.9	61.5
2-epoch compute control	control	48.6	[40.5, 56.8]	39.3	48.9
std prediction reward	reward	0.7	[0.1, 3.9]	0.0	1.0
mean-only rescue	reward	52.9	[44.7, 61.0]	40.7	51.6
mean-only, no signal	control	57.9	[49.6, 65.8]	44.3	52.6
all-fail-group filter	reward	40.7	[32.9, 49.0]	35.7	39.6
decoupled (capped) reward	reward	27.9	[21.1, 35.8]	30.0	31.2
Δ-accuracy reward	reward	32.1	[24.9, 40.2]	29.3	28.1
anchor-QA (recall) reward	reward	44.3	[36.3, 52.6]	33.6	44.3
self-report reward	reward	38.6	[30.9, 46.9]	39.3	49.5
8B					
baseline	—	45.0	[37.0, 53.3]	39.3	32.8
aux loss, gold (s0, locked)	loss	0.0	[0.0, 2.7]	0.0	0.0
aux loss, gold (s42, escaped)	loss	**85.0**	[78.2, 90.0]	72.1	72.4
aux loss, shuffled	loss	79.3	[71.9, 85.2]	66.4	78.6
prompt-only	—	52.1	[43.9, 60.2]	44.3	47.4
std prediction reward	reward	0.0	[0.0, 2.7]	0.0	0.0
4B, late-finishing					
std prediction reward (s42)	reward	0.0	[0.0, 2.7]	0.0	0.0
decoupled normalization (s42)	reward	32.1	[24.9, 40.2]	26.4	42.2

Table 4: Unified formal evaluation (E01): final checkpoints, 140-game `valid_seen`, sampled decoding at temperature 0.4, Wilson 95% intervals (computed by the evaluation driver from the logged three-decimal success rate at $n{=}140$; recomputing from raw counts can shift a bound's last digit by 0.1pt). Intervals are within-arm binomial only; they do *not* include seed variance, which the multi-seed arms show spans 4–17 points peak-to-peak among the healthy 4B arms (and 85 points in the bistable 8B gold family); single-seed rows (including both table-topping placebos) carry that additional cross-seed uncertainty (§6.6). With 29 arms, uncorrected pairwise comparisons invite multiplicity concerns; the categorical statements in the text rest only on gaps of ≥ 25 points with non-overlapping intervals, which survive any standard correction, and fine-grained rankings among adjacent arms are deliberately not interpreted. Training-time last-6 means shown for continuity with the curves in the text; prose retains last-6 numbers where it describes training dynamics. The unseen column is the full 134-game `valid_unseen` split evaluated under the same fixed-batch protocol, which pads it to 140 episodes (six games run twice); percentages and intervals are over the 140 episodes, so the upper bound at zero is 2.7%. Formal numbers for the estimator controls, the 8B gold s96 and half-weight arms, the Δacc second seed, and the remaining placebo seed replications (shuffled s96; random-vocabulary s42/s96) were pending at the data freeze; other late-finishing family arms are included above. The anchor-QA arm's training last-6 and formal endpoint coincide at 44.3 by arithmetic accident; the underlying counts differ.

6.5 Practical guidance

In brief (full prescriptions with numbers in Appendix M): (1) do not mix difference-form self-prediction shaping into GRPO's std-normalized reward when within-group task-return variance is small relative to the shaping term; (2) if reward-channel shaping is unavoidable, use mean-only normalization (decoupling under-performs; annealing rescues only inside the honeymoon window); (3) prefer signals whose within-group variance decays, and verify the decay empirically (Δacc is the cautionary test); (4) for the gain, add an auxiliary CE pass *as a regularizer*: any well-formed target works, do not pay for gold supervision, and treat scale as hazardous (at 8B run a seed pair and consider halving the weight); (5) monitor the joint precursor, never entropy alone; (6) read feature-set difficulty off the prediction-saturation form before spending compute.

6.6 LIMITATIONS

Seeds and statistics. The headline auxiliary-loss arm, its two leading placebos, and the ALFWorld baseline carry three seeds; the WebShop pair, decoupled, and Δacc carry two; most mechanism arms are single-seed, so cross-arm gaps inherit partially quantified seed variance (healthy-4B spread 4–17 points peak-to-peak; 85 in the bistable 8B gold family, whose six occupancy samples support a structural description, not a rate estimate). Every arm entering a cross-arm ordering claim has a formal 140-game endpoint and the headline gaps survive it; the mechanism battery and the caption's pending cells keep training-caliber endpoints only, and the out-of-distribution pass (ALFWorld only) preserves every headline structure while magnitudes compress (§6.4).

Scope. Three environments carry results (ALFWorld, WebShop, HiddenRule-Gym); a fourth, ScienceWorld, appears in the arm inventory (Appendix F) as in-flight arms only (one running, one queued) and contributes no interpreted result. The WebShop shuffled placebo (70.3) and the mean-only rescue of the instant-suppression form (59.4) are completed, so attribution and prescription carry cross-environment evidence. Collapse-timing observations rest on three scale points, and HRG conclusions are capacity-limited at 1.7B (the 4B capacity gate shows lift-off without rule learning). The 8B baseline anomaly is resolved by the unified evaluation (the training-time 32.8% evaluates at 45.0% over 140 games, inside the 4B baseline seed band), so scale-ordering claims are stated on formal numbers only (§6.4). All results are on a single base-model family (Qwen3, three sizes) with one agent framework and one prediction-signal design (the inventory's Qwen2.5-1.5B row is an early full-budget pipeline anchor, not a comparison arm); the mechanism is architecture-agnostic algebra, but its empirical constants (collapse timing, honeymoon peaks, gain magnitudes) should be read as Qwen3-specific until replicated elsewhere. Estimator scope is four cells wide: GiGPO reproduces the collapse, PPO/GAE and RLOO survive the identical recipe, and collapse aligns with the within-group σ division across all four (§6); the last cell, global-batch whitening (REINFORCE++), completed after the main data freeze and survives the identical recipe (48.4% last-6 against its own 55.2% baseline arm, no collapse window, no terminal triple), localizing the warning to the within-group σ division rather than std normalization per se (training-caliber, single-seed, post-freeze: reported as a boundary note, not entered into the formal ordering tables).

Resolved since the previous version. The small-group-$\hat{\sigma}$ collinearity concern and the single-environment concern were both closed by preregistered ablations (group sizes $8/16$; WebShop replication); detail in Appendix K.

7 CONCLUSION

Dense prediction rewards are a natural and popular remedy for sparse-reward LLM agents, and under group-normalized RL they destroy the policy: at every scale, group size, and fixed shaping coefficient we tested, std normalization converts a bounded, telescoping, verifier-graded signal into full-scale pressure toward predictability, and the optimizer builds a dark room. The mechanism is a one-line algebraic fact with three-point empirical confirmation; the danger is governed by the signal's within-group variance trajectory; the only interventions that work are structural and preventive; and when the same signal does help, through an auxiliary loss, placebo controls trace the gain to the update rather than the world model. Both halves come with boundaries. The reward-channel failure and the content-free attribution replicate on a second environment, where task difficulty selects the failure's form. At 8B the auxiliary channel becomes seed-determined: gold full-weight locks two of three seeds and sends the third to the study's best gold-signal result, while every content-free or reduced-weight arm stays healthy. Scale moves the randomness from collapse timing to collapse occurrence, and moves content from inert to risk-implicated: the gain is content-free everywhere, but the one 8B configuration that locks is the true-gold signal at full weight (a structural reading from a six-run family, §6.6). We report the mechanism, the criterion, the early-warning signature, and the ledger of preregistered predictions so the next design iteration of dense supervision in agent RL can be checked against them.

Reproducibility Statement

All experiments use a single aligned configuration across scales (train batch $8 \times$ group 4, validation 32 episodes at temperature 0.4 (16 for HiddenRule-Gym at 1.7B; the 4B HRG capacity arm used 32), seed 0 unless stated; Appendix C). Training-dynamics numbers are last-6 validation means from training-time evaluation; formal endpoint numbers come from the unified 140-game evaluation of every arm entering a cross-arm ordering claim (§6.4, Table 4); the mechanism battery keeps training-caliber endpoints (§6.6). Seed replication (seeds 42/96) covers the headline auxiliary-loss arm, its two leading placebo controls, the ALFWorld baseline, and the 8B gold arm, with second seeds also on the decoupled, Δacc, and WebShop auxiliary arms; the random-token placebo and the 8B shuffled, prompt-only, and half-weight arms are single-seed (marked in Table 4 and Appendix F); Appendix F inventories all 74 training arms of the study with their endpoints; both evaluation splits (140-game seen, 134-game unseen) are complete for every arm in Table 4, and formal evaluation of late-finishing arms was pending at freeze time. The agent framework builds on the open-source verl-agent stack. Our code is available at `https://github.com/RobertWangWang/verl-agent` (a fork of `https://github.com/langfengQ/verl-agent`): the reward pipeline, the auxiliary-CE channel with all placebo modes, HiddenRule-Gym, the WebShop and ScienceWorld service harnesses, run recipes, unit tests, and the preregistration digest file whose git history timestamps every registered prediction.

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A Registered Predictions Ledger

Kept verbatim with outcomes. Each preregistered prediction was committed as a full 64-character SHA256 digest of its plaintext to a version-controlled file *before* the corresponding run completed; the digest file and all plaintexts ship with the code release, so every "registered before outcome" claim below is independently checkable.

Table 5: Registered predictions and outcomes (excerpt: the most informative entries; the full 25-entry hash registry, which also covers the estimator, WebShop, ScienceWorld, and 8B-family arms, ships with the code release).

registered prediction	outcome	note
Dose–response: weaker collapses earlier, 8B latest	falsified $\rightarrow$ revised	replaced by saturation-race law (val-zero: 1.7B@30 $<$ 8B@65 $<$ 4B@85)
1.7B collapse at $\sim$step 30	hit	
Anneal arm: no recovery after $\lambda \rightarrow 0$	hit	irreversibility evidence #2; the turn precedes the anneal's tail ($\lambda \approx 0.068$ at the turn)
Rescue arm: no collapse without std	hit	causal localization
Arm D: upper-bound Φ hard to learn	hit	base-rate plateau 0.90–0.93

registered prediction	outcome	note
Arm D: prediction converts to success	not testable at training caliber; revised to partial hit by formal eval	prediction did not exceed base rate in training; the 140-game endpoint shows partial behavioral conversion (24.3% vs. 10.7% no-signal) and the probe $+0.16$ internalized knowledge (App. D)
Self-report: confidence saturates day one	hit	variance-trajectory retro instance
Self-report: harmless under std (variance $\to$0)	hit at training caliber; revised by formal eval	training last-6 49.5% = baseline; the 140-game endpoint revises the arm to 38.6%, below baseline, the table's largest downward revision (§6.4)
Anchor-QA: harmless, no gain (revised pre-launch)	premise falsified; formal endpoint level with baseline	recall did not saturate (0.65$\to$0.95); the formal endpoint (44.3%) lands level with the 44.1% baseline mean, and drag appears only at training caliber
Δacc progress reward: safe under std	premise falsified; outcome consistent with trajectory form	accuracy plateaued at 0.92, variance persisted, chronic drag (28.1%); endpoint form falsified prospectively a second time
Group-size ablation (n=8, constant budget): collapse delayed but not prevented	hit	literal zero at step 135 vs. 85; all-fail fraction climbs from an early $\sim$0.5–0.6 (rolling) to a terminal lock at 1.00, measuring the feedback loop directly
Group-size ablation (n=16): collapse further delayed	half-hit	collapsed (✓) at step 80, *earlier* than n=4; the two-point delay trend was falsified by the third point
λ-sweep ($\lambda \in \{0.01, 0.03, 0.1\}$): std runs behave identically	hit (mechanism)	advantage scales $\pm7.3/\pm9.1/\pm7.2$, no trend over tenfold λ; all collapse; timing within variance band
All-fail-group filter: rescues without touching std	hit	0% $\to$ 39.6%; partial recovery (-9.9 vs. baseline), consistent with discarding 23–77% of samples
Aux-loss gain replicates across seeds	hit at two of three seeds	matched-seed pairing $+19.8/+24.4/-1.6$ at seeds 0/42/96 (vs. the three-seed baseline mean: $+22.6/+16.8/+3.3$); magnitude varies, and the third seed lands at its own baseline
Random-vocabulary strong placebo: gains like gold/shuffled $\Rightarrow$ pure update/format effect; baseline band $\Rightarrow$ environment statistics are the active ingredient	hit (first branch)	73.4%, statistically tied with gold (69.3) and shuffled (76.0); attribution resolved to the update/format effect
R46a (compute-matched replay, no aux target): no gain	hit	48.9% $\approx$ baseline; compute/update count contributes nothing
R46b (random-token placebo, format-stripped): between baseline and random-vocabulary placebo	hit	61.5%, dead-center of the preregistered band; graded attribution ladder complete
Rescue from inside the absorbing state (mean-only switch): no recovery	hit	11 checkpoints at literal zero over a 50-step window; advantage scale confirms amplifier removed; the absorbing state is self-sustaining
Fast anneal completed inside the honeymoon window: avoids collapse	hit	40.1%; timed withdrawal is the schedule-side preventive route
Matched-variance noise reward: collapses, honeymoon absent	half-hit	no honeymoon (✓), severe drag -26pt (✓), but no absorbing state: collapse requires variance $\times$ hackability; criterion refined to two dimensions

B Additional Figures

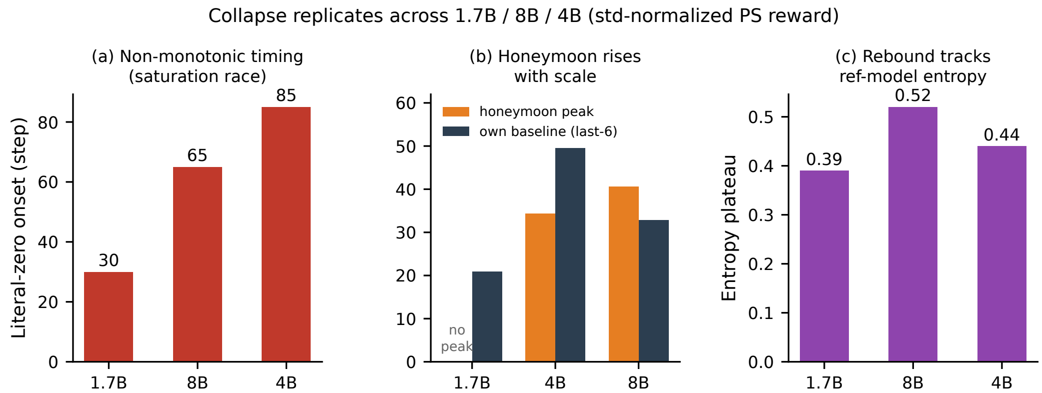

Figure 11: Three-scale summary: (a) non-monotonic literal-zero onset; (b) honeymoon peaks rising with scale (baseline bars are training-caliber last-6; the 8B formal baseline is 45.0, §6.4, which no peak exceeds); (c) post-collapse entropy plateaus track reference-model entropy.

C Reproducibility Details

train batch × group size	8 × 4 (= 32 trajectories/step)
validation	32 episodes (ALFWorld/WebShop), 16 (HiddenRule-Gym at 1.7B; the 4B HRG capacity arm used 32); sampled decoding, temperature 0.4, every 5 steps
horizon / max steps	50 (ALFWorld), 30 (HiddenRule-Gym), 15 (WebShop)
optimizer / lr	AdamW, 10^{-6}, KL loss coef 0.01 (low-var estimator)
lengths	prompt 1536 / response 512 tokens
task reward	terminal; ALFWorld/HiddenRule-Gym binary success, WebShop continuous matching score in $[0, 1]$
shaping	$\lambda = 0.1$, difference-form (superficially PBRS), injected per step-sample
advantage normalization unit	GRPO mean/std pooled over all step-samples of a group (cross-step pooling, the framework default; pool = 4 trajectories × $\leq$50 steps), z-score broadcast to each sample's response tokens (§3.2)
invalid-action penalty	0.1
auxiliary CE weight (loss-channel arms)	$\beta = 0.1$ (full weight; the 8B half-weight arm uses $\beta = 0.05$)
training length	150 steps (≈ 18–24 h per run; the post-hoc rescue extension runs from the step-125 checkpoint to step 175)
precision	FP32 master weights; BF16 compute (FSDP mixed precision, FP32 gradient reduce); BF16 rollout (vLLM). Critic (PPO/GAE arm) follows the actor scheme
hardware	8×96 GB GPUs per arm (4B/8B); 2×32 GB (1.7B, HRG)

Table 6: Shared configuration across all comparison arms (seed 0).

D Published reference points on these benchmarks

For context only: recent RL-trained agent results on the same benchmarks, with numbers quoted verbatim from the cited papers. None is comparable to our arms. They use different base models (Qwen2.5-Instruct family), full training budgets (ours run at one quarter budget by design, §4), and their own evaluation protocols and splits; we include them to locate the setup in the literature, not to rank against it. All of our claims remain same-budget arm-to-arm comparisons.

method	model	reported result	benchmark(s)
GRPO (as reported in Feng et al., 2025; He et al., 2026)	Qwen2.5-1.5B / 7B	$\sim$73–74% / $\sim$79–83%	ALFWorld
GiGPO (Feng et al., 2025)	Qwen2.5-1.5B–7B	>+12 / >+9 over GRPO	ALFWorld / WebShop
HCAPO (Tan et al., 2026)	Qwen2.5-7B	+13.8 / +7.7 over GRPO	ALFWorld / WebShop
SCPO (Xu et al., 2026)	Qwen2.5-1.5B	$93.7 \pm 4.1\%$ / $74.8 \pm 2.0\%$	ALFWorld / WebShop
BEACON (Wang et al., 2026b)	—	92.9%, long-horizon split (their GRPO: 53.5%)	ALFWorld; also WebShop, ScienceWorld
RLVMR (Zhang et al., 2025b)	7B	83.6%, hardest unseen split	ALFWorld, ScienceWorld

Table 7: Published reference points (quoted from the cited papers' reported results; protocols, budgets, and splits differ from ours and from each other).

E Additional Experimental Detail

Boundary conditions of the rescue. Std normalization is genuinely useful where it was designed: an adaptive gradient with provable convergence benefits when within-group reward variance is task-informative (Ge et al., 2026). Our results delimit the failure regime (a dense policy-dependent shaping term whose spread dominates the within-group variance; sparse terminal success with all-fail groups is the extreme case), and independent evidence that std removal repairs GRPO in stochastic-outcome domains (Bereket & Leskovec, 2025) suggests the repair generalizes along the axis of "variance not aligned with task quality." A learned critic derives its scale from bootstrapped value estimates rather than the spread of a small same-prompt group, so the all-fail degeneracy behind Proposition 1 has no direct analogue there; the completed PPO/GAE arm confirms it (identical recipe, 150 steps, no collapse window, 37.5%, prediction accuracy peaking at 0.60), and the RLOO pair replicates the pattern inside the group-relative family (40.6% vs. its own 44.8% baseline). The danger localizes to the within-group σ division, not group-relative estimation as such (§6).

Post-hoc belief probes (HRG). Privileged-latent probes on final checkpoints (leakage-audited questions; yes/no cells reported against majority-class baselines) give the belief-side view. State-memory probe accuracy orders the arms C=0.48 (0.61) $>$ task-weighted (0.46) $>$ pure GRPO (0.40) $>$ C=0.90 $\approx$ Arm D (0.35) $>$ generic-PS (0.31) against a 0.14 random floor: the training-time F1 finding (generic saturated signal erodes beliefs, task-relevant weighting preserves them) is reproduced by an independent instrument. Every arm except the mid-coverage and task-weighted ones sits at or below the no-signal 0.40, including the privileged-feature arm D (0.35), not only the two pathological regimes (saturation 0.31, overload 0.35): a non-covering, overloaded, or narrowly targeted Φ erodes general state memory, while mid-coverage and task-weighted Φ strengthen it (0.61, 0.46). Arm D's low general-memory score coexists with its +0.16 on the specific latent it was trained on (below): the cost is paid in breadth, not on the trained target. Arm D adds a properly-baselined dissociation: on the latent it was trained to predict it probes at 0.99 against a 0.83 majority-class baseline (a real +0.16 of internalized knowledge; this post-hoc yes/no probe is a different instrument from the training-time generative prediction accuracy, whose 0.90–0.93 plateau in §6.3 is scored against the environment, hence the different baselines), while task success converts only partially (24.3% vs. 10.7% at the 140-game endpoint, §6.4; the 16-game window had read it as floor). Knowledge internalization and behavioral conversion remain separable in degree, not absolutely. No arm exceeds the majority baseline on rule-relevance probes, consistent with none cracking the rules.

F Complete experiment inventory

All 74 training arms of the study, grouped by environment. Design-factor columns: σ = group-std normalization in the advantage; R = dense prediction signal through the *reward* channel; L = the same signal through the *loss* channel (auxiliary CE); $\varnothing$ = content-free variant (shuffled / random / noise). "last-6" is the training-time endpoint; "seen"/"unseen" are the formal endpoint evaluations of §6.4 where available (ALFWorld: 140-game seen and 134-game unseen padded to 140 episodes; 200 goals for WebShop; 140 probe-family games for HiddenRule-Gym). All completed arms ran the full 150-step budget under the aligned configuration (Appendix C), with one exception: the post-hoc rescue arm extends its parent run beyond the budget, a 50-step mean-only window from the step-125 checkpoint to step 175. Every substantive arm carried a hash-registered prediction (25 registry entries, several covering multi-arm batches, including the estimator, WebShop, ScienceWorld, and 8B-family arms; the registry file ships with the code release). Appendix A reproduces the most informative subset with outcomes. Arms marked (running)/(queued) belong to the frozen final queue. Within σ✓R✓ rows, outcome differences trace to the modifier named in the Configuration column (group filtering, anneal schedule, decoupled or mean-only normalization): the σ column marks presence of the within-group divisor, not absence of every mitigation.

Table 8: ALFWorld, Qwen3-4B (main battery)

		signal				Metrics (%)			
Configuration	σ	R	L	∅	seed	last-6$^{(\uparrow)}$	seen$^{(\uparrow)}$	unseen$^{(\uparrow)}$	outcome
Baselines and controls									
GRPO baseline	✓				0	49.5	42.9	42.1	baseline
GRPO baseline	✓				42	39.1	40.7	37.1	baseline
GRPO baseline	✓				96	51.6	48.6	46.4	baseline
mean-only control					0	52.6	57.9	44.3	control
2-epoch replay (compute)	✓				0	48.9	48.6	39.3	parity
Reward channel, std normalization on									
PS prediction reward	✓	✓			0	1.0	0.7	0.0	**collapse**
PS prediction reward	✓	✓			42	0.0	0.0	0.0	**collapse**
PS, λ=0.01	✓	✓			0	1.0	—	—	**collapse**
PS, λ=0.03	✓	✓			0	0.0	—	—	**collapse**
PS, group n=8	✓	✓			0	2.6	—	—	**collapse**
PS, group n=16	✓	✓			0	0.0	—	—	**collapse**
PS, slow anneal $\lambda\to 0$	✓	✓			0	0.0	—	—	**collapse**
PS + fast anneal	✓	✓			0	40.1	—	—	**rescue**
PS, GiGPO (step groups)	✓	✓			0	0.0	—	—	**collapse**
matched-variance noise	✓	✓		✓	0	23.4	—	—	severe drag
Δacc progress reward	✓	✓			0	28.1	32.1	29.3	chronic drag
Δacc progress reward	✓	✓			42	34.4	—	—	chronic drag
anchor-QA reward	✓	✓			0	44.3	44.3	33.6	training-only drag
self-report reward	✓	✓			0	49.5	38.6	39.3	parity/drag
Reward channel, normalization modified									
PS + mean-only		✓			0	51.6	52.9	40.7	**rescue**
PS + all-fail filter	✓	✓			0	39.6	40.7	35.7	partial rescue
PS + decoupled norm	✓	✓			0	31.2	27.9	30.0	drag
PS + decoupled norm	✓	✓			42	42.2	32.1	26.4	drag
mean-only@step125 (post hoc)		✓			42	0.0	—	—	no recovery
Loss channel (auxiliary CE)									
aux gold	✓		✓		0	69.3	68.6	65.0	gain
aux gold	✓		✓		42	63.5	57.9	45.0	gain
aux gold	✓		✓		96	50.0	52.1	39.3	parity
aux shuffled	✓		✓	✓	0	76.0	81.4	73.6	gain
aux shuffled	✓		✓	✓	42	60.4	67.9	66.4	gain
aux shuffled	✓		✓	✓	96	48.9	—	—	parity
aux random-vocab	✓		✓	✓	0	73.4	**87.1**	71.4	gain
aux random-vocab	✓		✓	✓	42	77.6	—	—	gain
aux random-vocab	✓		✓	✓	96	60.9	—	—	gain
aux random-tokens	✓		✓	✓	0	61.5	67.9	57.9	gain
aux noise (R60a)	✓		✓	✓	0	43.2	—	—	no gain
aux half-gold (R60b)	✓		✓	✓	0	(running)	—	—	
Estimator family (same PS recipe)									
PPO/GAE + PS		✓			0	37.5	—	—	survives
RLOO baseline					0	44.8	—	—	baseline
RLOO + PS		✓			0	40.6	—	—	survives
R++ baseline (E11a)					0	55.2	—	—	baseline
R++ + PS (E11b)		✓			0	48.4	—	—	no collapse (post-freeze)

Table 9: ALFWorld, Qwen3-8B (bistability family)

		signal				Metrics (%)			
Configuration	σ	R	L	∅	seed	last-6$^{(\uparrow)}$	seen$^{(\uparrow)}$	unseen$^{(\uparrow)}$	outcome
baseline	✓				0	32.8	45.0	39.3	baseline
PS prediction reward	✓	✓			0	0.0	0.0	0.0	**collapse**
aux gold	✓		✓		0	0.0	0.0	0.0	**absorbing**
aux gold	✓		✓		42	72.4	**85.0**	72.1	**best gain**
aux gold	✓		✓		96	0.0	—	—	**absorbing**

		signal				Metrics (%)			
Configuration	σ	R	L	∅	seed	last-6$^{(\uparrow)}$	seen$^{(\uparrow)}$	unseen$^{(\uparrow)}$	outcome
aux gold, β=0.05	✓		✓		0	65.1	—	—	escape+gain
aux shuffled	✓		✓	✓	0	78.6	79.3	66.4	gain
prompt-only	✓				0	47.4	52.1	44.3	level

Table 10: ALFWorld, Qwen3-1.7B / Qwen2.5-1.5B

		signal				Metrics (%)			
Configuration	σ	R	L	∅	seed	last-6$^{(\uparrow)}$	seen$^{(\uparrow)}$	unseen$^{(\uparrow)}$	outcome
1.7B baseline	✓				0	20.9	17.9	18.6	baseline
1.7B PS reward	✓	✓			0	0.0	0.0	0.0	**collapse**
1.5B full budget (anchor)	✓				0	67.2	—	—	anchor

Table 11: WebShop, Qwen3-4B

		signal				Metrics (%)			
Configuration	σ	R	L	∅	seed	last-6$^{(\uparrow)}$	seen$^{(\uparrow)}$	unseen$^{(\uparrow)}$	outcome
base, small	✓				0	60.4	60.0	—	baseline
PS reward, small	✓	✓			0	1.0	0.5	—	**instant collapse**
base, full	✓				0	29.2	28.5	—	baseline
PS reward, full	✓	✓			0	0.5	0.5	—	**collapse**
aux gold, small	✓		✓		0	71.4	63.0	—	gain→tie
aux gold, small	✓		✓		42	50.5	46.0	—	below base
aux shuffled, small	✓		✓	✓	0	70.3	**67.0**	—	gain
PS + mean-only, small		✓			0	59.4	57.5	—	**rescue**

Table 12: HiddenRule-Gym, Qwen3-1.7B/4B

		signal				Metrics (%)			
Configuration	σ	R	L	∅	seed	last-6$^{(\uparrow)}$	seen$^{(\uparrow)}$	unseen$^{(\uparrow)}$	outcome
pure GRPO, 1.7B	✓				0	8.3	10.7	—	floor
PS generic Φ	✓	✓			0	10.4	14.3	—	floor; F1 collapse
PS task-weighted Φ	✓	✓			0	5.2	8.6	—	floor; F1 held
arm D, privileged Φ		✓			0	10.4	24.3	—	partial conversion
coverage C=0.23		✓			0	12.5	26.4	—	gain
coverage C=0.48		✓			0	24.0	**34.3**	—	best HRG
coverage C=0.66		✓			0	14.6	13.6	—	mid dip
coverage C=0.86		✓			0	22.9	22.1	—	gain
coverage C=0.90		✓			0	3.1	10.7	—	benefit consumed
aux gold @C=0.48	✓		✓		0	14.6	17.1	—	marginal
aux shuffled @C=0.48	✓		✓	✓	0	14.6	15.0	—	tied w/ gold
pure GRPO, 4B	✓				0	26.6	—	—	off floor

Table 13: ScienceWorld, Qwen3-4B

		signal				Metrics (%)			
Configuration	σ	R	L	∅	seed	last-6$^{(\uparrow)}$	seen$^{(\uparrow)}$	unseen$^{(\uparrow)}$	outcome
base (boil)	✓				0	(running)	—	—	
PS reward (boil)	✓	✓			0	(queued)	—	—	

G Extended related work

Group-normalized RL and its advantage pathologies. Group-relative policy optimization (Shao et al., 2024; DeepSeek-AI, 2025) normalizes each group's returns by their mean and standard deviation. Dr.GRPO (Liu et al., 2025) identified this estimator's std and length biases; we adopt its mean-only variant as an *instrument* and contribute the causal demonstration that in long-horizon sparse-success agents the std term alone separates catastrophic from benign. Bereket & Leskovec (2025) independently show that removing group-std normalization fixes GRPO-induced overconfidence for stochastic outcomes, while Ge et al. (2026) defend std normalization as an adaptive gradient on reasoning benchmarks; our results delimit where that defense breaks: phases where an ε-scale shaping term supplies the dominant within-group variance, with sparse-success all-fail groups as the extreme case. Related diagnoses include heterogeneous-group bias (Zhu et al., 2025), local-normalization bias (Hu et al., 2025), baseline-induced entropy hazards (Wu et al., 2025), mean-only versus mean-plus-variance calibration theory (Mroueh, 2025), and GDPO's decoupled multi-reward normalization (Liu et al., 2026), which our decoupled arm tests and finds insufficient under sustained dense pressure. A distinct line treats *degenerate* (all-same-outcome) groups as zero-signal dead zones to be pruned, replaced, or reweighted (Xu & Ding, 2025; Zhang et al., 2026; Yang et al., 2026; Zhang et al., 2025a). Our setting inverts that premise: once a shaping term is injected, all-fail groups are *full-signal* zones, their ε-scale shaping differences amplified to full advantage scale (Proposition 1). Remedies that merely skip or downweight degenerate groups remove the amplifier's fuel without removing the amplifier: our filter arm shows this rescues only partially (-9.9pt vs. baseline, §5.3), and by suppressing the symptom such remedies can mask the mechanism. Concurrent work formalizes the group-std term as the single quantity on which GRPO, Dr.GRPO, and DAPO differ (Bay & Yearick, 2026), consistent with our localization; we add what that term *does* when an injected signal is the only within-group variance. Qualitative precedents of the amplification exist ("illusory advantages" in text-to-image GRPO, Wang et al., 2025d; scale-equalizing per-component normalization, Ichihara et al., 2025); Proposition 1 is, to our knowledge, its first formalization for the sparse-success all-fail-group structure of multi-turn agents. Our infrastructure follows the step-independent multi-turn line (Feng et al., 2025; Zhou et al., 2024); successors refine step-level advantage estimation (He et al., 2026) while retaining the group-std normalizer, and our results concern that shared normalization machinery, not any particular grouping scheme.

Dense signals, shaping, and reward hacking in agent RL. Prediction-*error* seeking (Pathak et al., 2017) is famously vulnerable to the noisy-TV trap (Burda et al., 2019; Mavor-Parker et al., 2022); our failure is the mirror image, prediction-*accuracy* seeking parking in predictable corners (Friston et al., 2012; Baltieri & Buckley, 2019), a pathology one deep active-inference agent already exhibited by mastering a single repeated action (Champion et al., 2024). On the shaping side, Ng-style invariance (Ng et al., 1999) assumes a policy-independent potential; dynamic and learned potentials break it in known ways (Devlin & Kudenko, 2012; Harutyunyan et al., 2015; Behboudian et al., 2021; Forbes et al., 2024), but that line never considers the interaction with group normalization, where our damage concentrates. Dense process rewards are known to be hacking-prone (Cui et al., 2025b; Pan et al., 2022; Wang et al., 2026a); the standard mitigation is the progress principle, reward change rather than level (Setlur et al., 2025; Hou et al., 2025; Wang et al., 2025a; Xi et al., 2026). Our Δacc arm is its preregistered test, and its failure (accuracy plateaus below saturation, variance persists, chronic drag) sharpens the boundedness principle of Fu et al. (2025): return-level boundedness does not survive group z-scoring at the step level. RAGEN's "Echo Trap" (Wang et al., 2025e) and stability reports for dense turn-level rewards (Wang et al., 2025c) are near neighbors observed without a controlled dense-signal culprit.

Auxiliary objectives and world modeling for agents. From UNREAL (Jaderberg et al., 2017) to modern LLM agents, world-model signals delivered through the loss channel have a consistent track record (Shrivastava et al., 2026; Lu et al., 2026; Voelcker et al., 2024). Nor is the reward channel uniformly hostile: RWML (Yu et al., 2026) succeeds with an embedding-space gap reward on ALFWorld while warning that token-level next-state prediction "can lead to model collapse," a one-line anticipation of the failure we characterize, and VAGEN (Wang et al., 2025b) rewards world-model reasoning under a redesigned estimator; our variance-trajectory criterion must be, and is, consistent with these successes (§3.3). What this literature lacks is a controlled comparison holding the signal fixed while varying only the consumption mechanism; our matrix supplies it (§5.4), with a placebo ladder, a compute-matched control, and a gradient-interference probe. Relative to ECHO (Shrivastava et al., 2026), which established the auxiliary-CE recipe's effectiveness, we contribute the attribution: neither content–context pairing nor environment statistics nor extra compute is the active ingredient.

Training instability, seed variance, and scale. Seed-dependent instability of language-model fine-tuning is well documented, including runs that diverge partway through training (Dodge et al., 2020), with vanishing gradients as one driver (Mosbach et al., 2021). Our 8B bistability (§6) is not this phenomenon rebranded: two of three gold seeds enter the same early collapse window (the third locks later, from the lead), the failed basin is a mechanistically characterized absorbing state (prediction-format saturation under task-gradient starvation), and the escaped basin lands at the study's best gold-signal result, a pattern consistent with two distinct attractors rather than diffuse variance, though with six occupancy samples it remains a structural hypothesis (§6.6). Scale-dependent instabilities invisible at smaller scales, and their predictability from internal signals, are documented for pretraining (Wortsman et al., 2023); our entropy-based precursor plays the analogous role for agent RL. Finally, the auxiliary-update framing connects to negative transfer in auxiliary-task learning (Jiang et al., 2023): at 4B the auxiliary CE pass is pure positive transfer, at 8B its sign becomes seed-dependent, a scale boundary auxiliary-task weighting schemes may need to negotiate.

H Collapse mechanism: measurement detail

Budget vs. published ALFWorld numbers. Published full-budget GRPO baselines on this stack reach $\sim$73–74% (1.5B) and $\sim$79–83% (7B) on ALFWorld (reported as the GRPO baseline in Feng et al., 2025; He et al., 2026). Our arms deliberately run at one quarter of that budget (32 trajectories/step, 150 steps) so that fourteen mutually comparable arms fit a fixed compute envelope; every claim is a same-budget arm-to-arm comparison, none depends on absolute success rates, and a full-budget sanity anchor (Qwen2.5-1.5B, 128 trajectories/step) reaches 67.2% in our hands, on the published baseline's convergence trajectory. Published reference points on these benchmarks, including current state-of-the-art claims, are collected in Appendix D with the protocol caveats that make them non-comparable to our same-budget arms.

A replay of a collapsed 1.7B checkpoint against a healthy same-scale policy (12 episodes each, identical harness) confirms the drift directly: state-visitation entropy drops tenfold (0.91→0.09; 1.25 locations per episode vs. 4.6), consecutive-action repetition rises 0.21→0.93, episode length pins at the horizon, and the collapsed policy is *more* compliant than the healthy one (invalid-action rate 0.005 vs. 0.173, prediction-parse validity 1.00): better behaved on every surface metric we log, and it solves nothing. Three direct measurements complete the picture. First, preregistered constant-budget group-size ablations (4×8 and 2×16) test whether the pathology is an artifact of unstable small-group $\hat{\sigma}$ at $n{=}4$: collapse occurs at every group size with *non-monotone* timing (literal zero at steps 85/135/80), the $n{=}8$ delay a two-point comparison would read as a trend being overturned by the third point; timing is dominated by dynamics variance, not $\hat{\sigma}$ stability. Second, these runs log the all-fail group fraction: it climbs and locks at 1.00 (every group all-fail, task gradient identically zero, updates driven *entirely* by the prediction reward), the lock coinciding with the literal-zero window in both measured runs (Figure 3); a calibrated retrodiction of the original collapse run shows the same ramp ($\sim$0.65 $\rightarrow$ $\geq$0.9), while the rescued and baseline arms' fractions instead decay to $\sim$0.25 as success accumulates. Third, the coefficient is empirically irrelevant inside all-fail groups, as Proposition 1 predicts: a tenfold reduction leaves the amplified scale unchanged ($\pm$7.2 at $\lambda{=}0.1$ vs. $\pm$7.3 at 0.01) and the arm collapses on schedule (last-6 1.0%); annealing has nothing to act on.

Early-warning signature. The joint precursor is entropy monotone decline $\wedge$ prediction saturation $\wedge$ length pinning, leading the turn by 15–30 steps. Entropy alone false-alarms three ways (victory sharpening: the rescued arm dips to 0.095 while winning; early sharpening: the decoupled arm dips to 0.094 with prediction *declining*; pre-breakthrough consolidation: the auxiliary-loss arm reaches 0.04 right before its 28→59% phase transition), and it also declines for reward-independent reasons under clipped policy-gradient training (Park et al., 2025a), so the conjunction is required. We froze the rule as entropy-has-declined $\wedge$ prediction $\geq$0.95 $\wedge$ length $\geq 0.95\,H$, evaluated over history since the components arrive sequentially. Its thresholds were calibrated only on the four definition-set collapse runs (the three-scale collapse arms and the slow-anneal arm); the conjunction requirement itself was motivated both by those retrodictions and by the entropy-only dips already observed in healthy runs. Evaluated once, at a fixed date, on the fourteen runs then available outside the calibration set,[2] it detects all three held-out *reward-channel* collapses, 73/53/13 steps ahead of the collapse point, defined as the first validation opening a sustained dead band ($\leq$6.5% with no recovery; the first *exactly*-zero checkpoint can lag it, e.g. step 135 vs. 130 for n=8, and the 15–30-step precursor lead above is measured to the still-earlier *turn*: three distinct calibers). Its scope is that failure class: the 8B auxiliary-loss collapses (two locked seeds; the direct response-length measurement is from seed 0) bypass the length-pinning component altogether (§6), structurally outside the rule's coverage rather than a miss. Among the eleven arms healthy at training caliber it raises one alarm, on the self-report arm, whose formal endpoint later revises below baseline (38.6%, §6.4); that cuts both ways (a true detection the 32-game window could not resolve, or a false positive on an arm that merely underperformed; the arm never collapses, so the variance criterion's no-collapse call stands), and we report the record conservatively: three of three detected, at most one false alarm.

I Fix hierarchy and signal-form detail

Irreversibility. Constant pressure: 120 post-collapse steps at fixed λ (1.7B), zero recovery. Annealing: cosine $\lambda \to 0$ still collapses, the turn arriving at $\lambda \approx 0.068$ (step $\sim$57 of the schedule; literal zero follows by $\sim$85), and its final $\sim$30 steps at $\lambda < 0.01$ (effectively pure GRPO) show zero recovery; entropy remelts to 0.439, matching the un-annealed arm's 0.44 (Fig. 4). Post-collapse entropy rebounds at all scales (1.7B 0.39, 4B 0.44, 8B 0.52; ordered by reference-model entropy, consistent with KL-anchored dynamics; Cui et al., 2025a): the absorbing state is *behavioral*, not temperature-level, and restoring entropy does not restore the policy, so fixes must be preventive. The preregistered rescue arm completes the intervention set: resuming the seed-42 collapsed arm from its step-125 checkpoint, 70 steps past that arm's first zero-success evaluation, and switching to mean-only normalization (the intervention that prevents collapse from the start) recovers *nothing* over a 50-step window, 11 validation checkpoints at literal 0%. The measured advantage scale confirms the amplifier is gone: magnitudes fall from $\pm$5–9 to $\approx$0.1, the raw per-step scale that λ and the invalid-action penalty share. Prediction accuracy sits saturated at 1.0, so the shaping term contributes almost no within-group spread; the residual $\approx$0.1 excursions are raw-scale outliers (consistent with the collapsed policy's rare invalid steps, rate 0.005), not amplified signal, and there is no task gradient to escape with: the absorbing state is self-sustaining. Late scheduling, coefficient scale, and post-hoc structural intervention all fail; three preventive routes work: structural (mean-only from the start), sample-level (dropping all-fail groups, partial rescue at 39.6%, §5.3), and *timed withdrawal* (a preregistered fast-anneal arm completing the same cosine schedule inside the honeymoon window avoids collapse entirely, 40.1%, Fig. 4). Irreversibility binds after the turn, not before.

Matched-variance noise isolates hackability. A preregistered control replaces the prediction score with random potentials of matched per-step variance: the amplifier stays fed, but the policy has no control over the signal. The arm shows no honeymoon, drags severely (23.4%, −26 vs. baseline), and yet never enters an absorbing state across 150 steps. Sustained variance is therefore sufficient for damage but not for the lock; the absorbing state additionally requires the signal to be *optimizable by the policy*. This is the second axis of the criterion (§3.3); the two axes together are necessary in every observed reward-channel lock but not jointly sufficient: the Δacc and anchor-QA arms satisfy both and neither locks (chronic drag for Δacc; training-caliber-only drag for anchor-QA, whose for-

[2]Three collapses: group-size n=8 (+73), the λ=0.01 arm (+53), and group-size n=16 (+13). Eleven training-healthy arms: the seed-0/42 baselines, mean-only PS and its no-signal control, anchor-QA, Δacc, the all-fail filter, the gold, shuffled-gold, and random-vocabulary auxiliary arms, and self-report. "Healthy" means no sustained zero-success validation window at training caliber, so drag arms count as healthy. Two of the healthy arms (mean-only PS, aux gold) supplied the pre-freeze entropy-dip observations that motivated the conjunction, so their clearing is confirmatory rather than independent; excluding them leaves nine independent healthy arms with the same single alarm. The decoupled arm, though completed by the evaluation date, is absent from the recorded roster, an omission we flag rather than justify; at its entropy dip its prediction accuracy was declining, below the rule's 0.95 gate, so the conjunction would not have alarmed on it, and its inclusion could only have enlarged the healthy denominator. Arms completing after the evaluation date, whether collapsing (λ=0.03, GiGPO, the 4B PS seed-42 arm, both WebShop arms) or healthy (matched-variance noise, fast anneal), were never scored against the rule; three-of-three is a claim over the evaluated set, not over all later collapses.

mal endpoint is level with baseline, §6.4). It is why the noise arm's registered prediction scored a half-hit (Appendix A).

Decoupling is not enough. Per-channel advantage decoupling (GDPO-style; prediction channel independently normalized, contribution capped at λ) does not collapse; the cap holds empirically (0.066–0.077). But it ends at **31.2%** last-6, well below the mean-normalized arms; a second seed lands level with its matched baseline at training caliber (42.2 vs. 39.1) but reproduces the deficit formally (27.9/32.1 against matched baselines 42.9/40.7; descriptive comparison). The measured scales carry the lesson (Fig. 5): the healthy mean-only arm's raw shaping advantage (± 0.098) is *larger* than the dragging decoupled arm's capped one (0.066–0.077), so magnitude alone does not order the outcomes; amplification structure does. The per-channel z-score re-stretches vanishing raw differences to a fixed λ-scale pressure regardless of the signal's actual spread, a capped copy of the same amplifier, whereas the mean-only contribution is the raw $\lambda(r - \bar{r})$, tied to that spread. The ladder reads: unamplified $\approx\lambda$ harmless / amplified-but-capped chronic drag / amplified full-scale lethal, ordered by structure, not magnitude.

J Attribution ladder detail

Two further controls complete a graded attribution ladder. The preregistered strong placebo (random-vocabulary: environment-disjoint words; SHA256 digest published before completion) reached **73.4%**, statistically tied with true gold (69.3%) and shuffled gold (76.0%); true gold is, if anything, the lowest of the three. A compute-matched replay control (double PPO epochs, no auxiliary target) gains nothing (48.9%), and a format-stripped placebo (CE on bare out-of-domain words, no schema) captures a majority of the effect (61.5%). The ladder reads: baseline 49.5 $\approx$ compute replay 48.9 $<$ random-token placebo 61.5 $<$ schema placebos 73.4–76.0 $\approx$ true gold 69.3. The active ingredient is the auxiliary CE pass on *novel target strings* ($\sim +12$ training caliber; $+25$ formal over the matched seed-0 baseline), with schema-consistent formatting adding $+7.8$ to $+14.5$ at training caliber that the formal ladder revises to anywhere from $+0.7$ (gold) to $+19.2$ (random-vocabulary) over the format-only rung: the decomposition's magnitudes are caliber-dependent and not cleanly additive, while the ordering (compute $\approx$ baseline $<$ format-only $\leq$ schema placebos) and the content-free conclusion are stable across calibers. Compute contributes nothing and content contributes nothing *positive* at any level (formally, gold sits below the placebo level at both matched seeds: 68.6 vs. the three-placebo s0 mean 78.8, and 57.9 vs. the single completed s42 placebo endpoint 67.9, §6.4). This is a regularization / format-anchoring effect, echoing in-context results where label-randomized demonstrations retain most of their benefit (Min et al., 2022). A preregistered gradient-noise falsifier closes the remaining alternative: replacing the auxiliary CE gradient with structure-matched, zero-mean per-token sign noise (Rademacher $\pm\beta$, everything else identical) lands at 43.2% last-6, inside the baseline band and 26 points below gold; arbitrary update perturbation of matched magnitude buys nothing, so the gain requires *directed* novel-target gradients, not update noise (training caliber; registered in the review batch, outside the fourteen-arm matrix). Stated at its strongest: on class means, a format-consistent auxiliary CE pass on *arbitrary* strings outperforms every reward-channel delivery of a genuinely informative signal (the weakest placebo seed, shuffled s96 at 48.9, lands level with the mean-only arms). This bears awkwardly on the auxiliary-world-model literature: gains attributed to world modeling may be substantially attributable to the auxiliary update itself, a hypothesis our controls isolate but only compute-matched ablations in those systems can settle.

K WebShop and HiddenRule-Gym detail

The reward-channel failure generalizes, and the task regime sets its *form*. Full-mode WebShop collapses after a parity phase (turn $\sim$60, sustained zero from $\sim$105, within the 30–135 val-zero range the ALFWorld arms span once the group-size ablations are included): the ALFWorld arc with the honeymoon replaced by baseline tracking, so not a honeymoon in the Table 1 sense. Small-mode WebShop shows *instant suppression*, completing a four-form taxonomy: honeymoon-collapse (ALFWorld), collapse-after-parity (full-mode WebShop, the same delayed arc with the honeymoon replaced by parity), instant suppression, and chronic drag (the capped arms). The regime split fits the two-condition account (§3.3): in the easy regime (baseline 60%) a continuous task score over short episodes lets prediction variance dominate the group advantage from step 0, while in the hard regime (baseline 29%) scores run low and within-group return spread is small, restoring the runway in which the shaping term comes to dominate, and with it the delayed collapse-after-parity arc (early dense rewards biasing learning away from exploration is a known developmental-RL effect, Park et al., 2025b). The environment moderates form and timing; occurrence tracks the channel. The attribution and the rescue both transfer: a shuffled-gold placebo lands at 70.3%, statistically tied with true gold's 71.4% (formally 67.0 vs. 63.0, the placebo again at or above gold), so the content-free result is not an ALFWorld artifact, and mean-only normalization applied to the instant-suppression regime lands at 59.4%, parity with the 60.4% baseline and a $+58$-point rescue over the std arm. (WebShop has no mean-only no-signal control, so the parity reading is a cross-normalization comparison like the HRG references; since the normalizer difference, $+3.1$ training / $+13.8$ formal on ALFWorld, could only flatter the arm, and it still does not beat the baseline, the no-gain regularity holds a fortiori.) A second auxiliary-loss seed qualifies the gain sharply: 50.5% training caliber, -9.9 below its baseline, 46.0 vs. 60.0 formal (-14.0); the sign itself flips with the seed, and the first seed's formal gain compresses to $+3.0$ (63.0 vs. 60.0), statistically level. What replicates across environments is the seed-to-seed bandwidth, not the gain's direction: the WebShop auxiliary result is a single-seed-pair difference ($+11.0$/-9.9 training; $+3.0$/-14.0 formal), not a confirmed effect size.

The loss-channel gain is not unconditional. On HiddenRule-Gym at 1.7B (coverage-0.48 protocol), the same auxiliary recipe reaches 17.1% under the 140-game endpoint protocol (§6.4): statistically inseparable from the no-signal GRPO arm (10.7%; a clean std-to-std comparison) and below the coverage-matched reward arm (34.3%, a 17.2-point gap with disjoint intervals, under the categorical bar of §6.4 and cross-normalization: the reward arm is mean-only while the auxiliary arm keeps std). The gain has conditions (capacity, coverage structure, environment class) that one environment success plus a seed-split second (WebShop, §6) do not license extrapolating away. Its shuffled pair lands at a statistically tied 15.0%: even this boundary-regime non-gain is content-free, so the attribution of §5.4 now holds in all three environments.

Does the failure depend on the advantage estimator? Four completed estimator arms (GiGPO; PPO/GAE; the two arms of an RLOO baseline–signal pair) align on a single design bit. GiGPO (Feng et al., 2025), which adds *step-level* grouping on top of episode-level groups while keeping the group-std normalizer, collapses on schedule (a healthy 19–34% band for $\sim$45 steps, then sustained literal zero with the familiar terminal signature: prediction accuracy 1.000, prediction-parse validity 1.000); finer grouping widens the supply of degenerate groups rather than fixing them. PPO/GAE (a learned critic, no group statistics) survives the identical recipe: 150 steps, no collapse window, prediction accuracy peaking at 0.60, a 37.5% endpoint. RLOO (a leave-one-out mean baseline: grouped, but no std division) survives likewise at 40.6% against its own baseline arm's 44.8%. Collapse therefore tracks whether the advantage *divides by a within-group σ*, not grouping granularity or critic versus group baseline; the signal (present in every cell except the RLOO baseline) is necessary for the failure (§5.3) but does not by itself decide it. Two qualifiers keep this honest: all estimator endpoints here are training-caliber (formal evaluations pending at the freeze), so the load-bearing contrast is collapse versus none, not the orderings (the PPO arm moreover has no matched PPO baseline, and the RLOO pair's -4.2 sits inside single-seed noise). The REINFORCE++ pair (global-batch whitening) was the last cell: its baseline arm completed healthy (55.2% last-6, the strongest baseline in the estimator family), and its signal arm, registered before launch and completing after the main data freeze, resolves the cell on the survival side: 48.4% last-6 with no collapse window at any point (series minimum 15.6%, no dead band), no terminal triple at step 150 (prediction accuracy 0.556, mean episode length 33.7 of 50, entropy 0.185), and an all-fail-group fraction of 0.375 rather than a lock at 1.0. Had it collapsed, the warning would have widened to all std normalization; its survival localizes the warning to the within-group σ division (training-caliber, single-seed, post-freeze; its -6.8 against the pair baseline sits inside the single-seed noise band and is not interpreted).

K.1 Separating coverage, dynamics, and capacity

HRG's exactly computable coverage lets us separate what ALFWorld conflates.

Coverage axis. Arm A (pure GRPO): statistically at the random floor (last-6 8.3%, formal 10.7%, against 7.7%, the empirical success rate of a uniform-random valid-action policy over 1,000 episodes) with *gradient starvation* (grad norm 0.05), the near-zero-difference pathology (residual within-group spread reduces to occasional invalid-action penalties, and most groups have none at all, placing this arm in the $\sigma_p \to 0$, ϵ-floor branch of Proposition 2) and the opposite pole from ε-amplification. Arm B (PS, generic Φ): prediction saturates 0.99 in ∼20 steps, success stays statistically at the no-signal level (formal 14.3 vs. 10.7, overlapping intervals), and the belief-probe F1 *collapses* 0.51→0.24. The reward does control which beliefs are maintained, but here it was pointed at non-covering content. (No absorbing state forms in this environment: with the baseline itself at the random floor, there is no success rate to destroy, so the absorbing-state phenotype cannot manifest; the shaping pressure surfaces instead as the belief-probe collapse above. This is why the abstract's collapse count excludes the floor-bound HRG arms.) Arm C (task-relevant reweighting within the same Φ family): F1 holds at 0.49, success still floor; reallocation within a non-covering family does not buy coverage. Arm D (upper-bound privileged feature, clean mean-only dynamics): prediction plateaus at the base-rate ceiling 0.90–0.93 without ever cracking the rule; the 16-game training window read its success as floor (10.4%), but the 140-game endpoint evaluation (§6.4) revises this to 24.3% against the 10.7% no-signal reference (disjoint intervals, though the 13.6-point gap sits under the categorical bar of §6.4, with the further caveat that the reference arm is std-normalized while arm D is mean-only, folding in a normalization difference that measures +3.1 training / +13.8 formal on ALFWorld and is therefore material at this gap size, Fig. 10): the internalized feature *does* convert partially, on a scale the small validation window could not resolve. Feature difficulty remains measurable from the saturation *form*: instant 0.99 means trivially predictable; a base-rate plateau means beyond capacity.

Capacity axis. The same pure-GRPO recipe at 4B lifts off the floor (26.6% vs. the 1.7B arm's 8.3%, $\approx$3.2×; training caliber, 32- vs. 16-episode windows; grad norm ∼1.0 vs. starvation 0.05 at 1.7B) yet plateaus for all 150 steps: capacity explains the 1.7B floor (necessary) but does not by itself crack hidden rules (not sufficient). Train success 0.19–0.34 without validation transfer indicates memorization rather than rule inference. This arm is outside the formal-evaluation roster, so the capacity reading is training-caliber only.

Family controls localize the lock. A *shuffled-gold* arm (targets randomly permuted across steps) enters no collapse window and climbs to **78.6%** (formal 79.3, statistically level with the gold escape seed's 85.0): the *gain* side of the content-free attribution carries to 8B. The *risk* side does not. A *prompt-only* control (predict block in the prompt, no training signal) lands at 47.4%, healthy throughout, so the prompt change alone neither collapses nor explains the locked run. Across the first four occupancy samples (gold s0: 0.0; gold s42: 72.4; shuffled: 78.6; prompt-only: 47.4) only one had locked, and the lock looked like a low-occupancy accident. Two further family arms complete the occupancy picture (Fig. 8) and overturn that reading. A half-weight arm (β=0.05, on the locked seed) escapes and finishes at 65.1%: halving the coefficient turns the locked absorbing state into an escapable dip, so the coefficient scale participates causally in basin entry. (No tension with Proposition 1: the coefficient cancelled there is the reward-channel λ inside a z-scored advantage; β weights a loss term that never passes through advantage normalization.) A third gold seed (96) answers bimodal-versus-continuous in the bimodal direction: it leads the field through step 40 (71.9%), then collapses to literal zero at ∼step 85. Gold full-weight at 8B locks two of three seeds; every content-free or reduced-weight arm stays healthy. The risk concentrates in the true-gold signal at full weight. Under a content-blind null at the observed gold lock rate (two of three), the two healthy arms that carry the auxiliary CE channel (shuffled, half-weight) have probability $(1/3)^2 \approx 0.11$; counting the prompt-only arm too, which carries no auxiliary CE and is arguably not lockable under the null, would give 0.04. Suggestive either way, but six samples remain six (§6.6).

L Formal evaluation: out-of-distribution pass and calibration

(ii) The content-free attribution is now the study's most consistent formal pattern, stated within scale: at 4B the shuffled-gold s0 placebo (81.4%) and random-vocabulary arm (87.1%) sit above every true-gold seed (68.6/57.9/52.1), while the shuffled s42 and random-token endpoints (67.9) do not. Across scales the highest *gold* endpoint is the 8B escape seed (85.0%; the table's overall maximum remains the random-vocabulary placebo), whose two sibling seeds lock at zero; at 8B, placebo-versus-gold ordering is replaced by the lottery structure of §6. The attribution ladder restates formally on matched seed-0 arms as 42.9 (baseline) $\approx$ 48.6 (compute control) $<$ 67.9 (random-token placebo) $\approx$ 68.6 (gold) $<$ 81.4–87.1 (schema placebos): the format-only rung already matches gold, and the schema placebos exceed it. One calibration note: the placebo–gold gaps themselves (12.8–18.5 points) fall below the caption's categorical threshold (the shuffled placebo's interval moreover overlaps gold's; the random-vocabulary interval clears gold's, but that row is single-seed and the gap is under 25 points), so what survives is a descriptive class-level ordering, stated seed-paired (the three-placebo s0 mean 78.8 vs. gold 68.6 at seed 0; the single completed s42 placebo endpoint 67.9 vs. 57.9 at seed 42; the pooled means 76.1 vs. 59.5 average four placebo endpoints, three s0 and one s42, against the three gold seeds), not a categorical claim; pointwise, two placebo endpoints (67.9) sit just below the strongest gold seed (68.6), so even per-seed placebo $\geq$ gold does not hold.

Out of distribution, every headline structure survives and every magnitude compresses. The five collapsed arms are literal zero there too (upper bounds 2.7%): the dark room is a property of the policy, not the evaluation set. The channel dichotomy holds (every reward-channel variant at or below the 41.9% unseen baseline mean; the mean-only pair stays tied, 40.7 vs. 44.3), and the content-free ordering is preserved (placebos 57.9–73.6 above the gold mean 49.8). Four small near-baseline gaps do flip sign against their matched seed-0 references (mean-only rescue $+10.0 \rightarrow -1.4$, compute control $+5.7 \rightarrow -2.8$, anchor-QA $+1.4 \rightarrow -8.5$, and gold s96 against its matched s96 baseline $+3.5 \rightarrow -7.1$), all with overlapping intervals; the claims here rest on the large gaps, not on those cells. What compresses is magnitude: the auxiliary gain over baseline goes from $+15.4$ to $+7.9$, the two later gold seeds falling into the baseline band; direction survives, size roughly halves, the seed-variance theme repeats. The largest single reversion is the random-vocabulary placebo ($87.1 \rightarrow 71.4$, ceding the top spot to shuffled-gold at 73.6): within-class *rankings* are seed-unstable even though the class-level claim (no placebo seed below the baseline band or collapsing; training-caliber class means 66.2 vs. gold 60.9) is seed-replicated.

The weakening: self-report drops to 38.6% (from a training-time 49.5), the table's largest downward revision, moving it from "parity" to "below the baseline mean at point estimate" (the interval still overlaps the baseline's). A single 32-game-window number moved 11 points under formal evaluation: that is the justification for this section, and the reason the single-seed rows above, including the table-topping placebos, carry the caption's cross-seed caveat until their seed replications complete.

Resolved since the previous version. The concern that group size n=4 is collinear with the std pathology (small groups give unstable $\hat{\sigma}$) was closed by preregistered constant-budget ablations at n=8 and n=16 (§5): collapse occurs at every group size with non-monotone timing (85/135/80), so the pathology is neither a small-group artifact nor reliably mitigated by larger groups. The single-environment concern was closed by the WebShop replication (§6).

M Extended practical guidance

1. Do not mix difference-form self-prediction shaping into GRPO's std-normalized reward when within-group task-return variance is small relative to the shaping term. The sparse-success all-fail regime is the extreme case (our collapse configuration), and the WebShop easy regime shows a second route (continuous scores over short episodes let prediction variance dominate the group advantage from step 0). The prescription is conditional: dense reward-channel supervision with different variance trajectories has succeeded on this very benchmark (§2). The environment changes the failure's *form* while leaving its occurrence intact (§6).
2. If reward-channel shaping is unavoidable, use mean-only normalization. Per-channel decoupling under-performs (formal 27.9/32.1 across its two seeds vs. the 52.9–57.9% mean-only pair, a 21–30-point deficit; -12 to -16pt vs. the baseline mean at point estimate), and annealing rescues only when timed early: completed *inside* the honeymoon window it prevents collapse (40.1%), while late annealing fails and no post-hoc rescue attempt succeeded (irreversibility, §5.2).
3. Prefer signals whose within-group variance decays over training, but verify empirically that they actually saturate in your environment. Our preregistered Δacc test (echoing the

progress principle of Setlur et al., 2025 and Hou et al., 2025) is the cautionary result: accuracy plateaued below saturation, progress variance persisted, and the arm ran 12 points below the baseline mean at formal point estimate (32.1% vs. 44.1%, intervals overlapping; a second seed reproduces the direction at training caliber, 34.4 vs. its matched 39.1). Constructions that force variance to zero at mastery do not bind if mastery is not reached.

4. If you want the gain, add an auxiliary CE pass *as a regularizer*, and know what you are buying. Any well-formed target works: under formal evaluation at 4B, content-free placebo targets as a class match or beat true-gold targets (seed-paired: the three-placebo s0 mean 78.8 vs. gold 68.6; the single completed s42 placebo endpoint 67.9 vs. 57.9; the single-seed random-vocabulary endpoint, 87.1%, tops the table). **Do not pay for gold process supervision, and do not expect the gain to scale with signal quality**: the content is inert for the gain (§5.4, §6.4), the gain comes from the update itself and not from update noise (a structure-matched sign-noise falsifier stays at the baseline band, §5.4), with the task reward untouched and zero measured gradient interference, and on WebShop the outcome is seed-split ($+11.0$/-9.9 training caliber; formal $+3.0$ statistically level and -14.0; hard-regime cell untested). Treat model scale as hazardous: at 8B the same recipe is bistable, the risk concentrating in true-gold targets at full weight (two of three gold seeds lock; shuffled, prompt-only, and half-weight arms all stay healthy; a structural reading from six occupancy samples, two of them content-free, §6.6). At scale, run a seed pair before trusting it, and consider halving the auxiliary weight; in our single test that turned the locked configuration into an escapable one (65.1%) (§6).
5. Monitor the joint precursor (entropy decline $\wedge$ prediction saturation $\wedge$ length pinning), never entropy alone (three false-alarm modes documented).
6. Check feature-set difficulty via the prediction-saturation form before spending compute: instant 0.99 = trivially predictable feature set; base-rate plateau = beyond model capacity.

Seeds and statistics. The headline auxiliary-loss arm carries three seeds (69.3/63.5/50.0: direction stable at seeds 0/42, magnitude -1.6 to $+24.4$ under matched-seed pairing), and the ALFWorld baseline carries three (49.5/39.1/51.6). The WebShop replication carries two (training 71.4/50.5; formal 63.0/46.0 against a 60.0 baseline), reproducing the seed-to-seed spread while the gain's sign flips ($+11.0$/-9.9 training, $+3.0$/-14.0 formal), so we report the cross-environment gain as a single-seed-pair difference, not a confirmed effect size. Most mechanism arms are single-seed; the decoupled and Δacc arms carry a second seed each (decoupled 31.2/42.2 last-6, an 11-point spread that is itself a measurement of this uncertainty, both formal endpoints below their matched baselines; Δacc reproduces the drag direction at training caliber), so cross-arm gaps inherit seed variance that is only partially quantified. The 8B family carries six occupancy samples (gold 0.0/72.4/0.0, shuffled 78.6, prompt-only 47.4, half-weight 65.1); the third gold seed answered bimodal-versus-continuous in the bimodal direction, and the layered pattern (gold full-weight locks two of three; every content-free or reduced-weight arm healthy) points at true gold at full weight as the risk concentration, but six points cannot estimate occupancy rates: the structural description is not a probability estimate. The 32-game validation limitation is now mostly closed: every completed arm's final checkpoint in Table 4 has a formal 140-game evaluation and the headline gaps survive it, while the mechanism battery (λ ladder, group sizes, anneal pair, noise, post-hoc rescue) and the cells the caption marks as pending (estimator controls, 8B gold s96 and half-weight, the Δacc second seed, late placebo seeds) keep training-caliber endpoints only (§6.4). The out-of-distribution pass (134-game unseen split; ALFWorld arms only, WebShop and HiddenRule-Gym have no unseen split) is complete: every headline structure is preserved and magnitudes compress (four near-baseline gaps flip sign within noise, §6.4), so distribution-shift claims rest on measured numbers.

N CRITERION: COMPATIBILITY AND IDENTIFIABILITY DETAIL

Whether a signal saturates fast enough to exit the danger window is empirical, and our anchor-QA arm is the cautionary instance: we predicted fast saturation, recall instead climbed slowly (0.65→0.95) and variance persisted; the arm showed mild drag at training caliber, though its formal endpoint lands level with the baseline mean (44.3 vs. 44.1, §6.4), so the instance supports the premise (saturation cannot be assumed) rather than the damage prediction. The other three signal forms then separate as the criterion predicts. Difference-form scoring keeps fluctuating per step as long as accuracy is high but short of saturation and the policy still visits varied states, so amplification persists through the entire approach to collapse; only inside the absorbing state, where accuracy pins at 1.0 and behavior degenerates to a fixed loop, does the variance reach zero (§5.2), when no task gradient is left to escape with. Saturated always-positive confidence becomes a constant: variance $\to 0$, the signal-fed amplification ends (what remains amplified is the penalty spread, which every healthy baseline also carries), no collapse (the self-report arm ran at baseline parity throughout training; its formal endpoint later revises low, a revision whose reading, true detection or measurement artifact, is left open in §5). Progress-style Δacc rewards have variance $\to 0$ at mastery *by construction*, but our preregistered test is the second cautionary instance. The arm's reward is $r_{\Delta\text{acc}} = \lambda\,\mathrm{clip}(\Phi_t - \Phi_{t-1}, 0, 1)$: the positive clipping already removes it from the difference-form family of §3.1, since a nonnegative reward's episode sum no longer telescopes to a bounded term. The running accuracy mean plateaued at 0.92 while the per-step score Φ_t kept fluctuating around it, so positive increments kept occurring: the unscaled clip term still paid a per-step mean of 0.095 at step 150 ($\lambda \times 0.095$ injected), variance persisted, and the arm ended at 28.1% (chronic drag, the same outcome rung as the decoupled arm); the endpoint form of the criterion would have called this arm safe, the trajectory form correctly does not. A matched-variance noise control adds the second axis: sustained variance supplies the pressure, but the absorbing state also requires the signal to be *hackable* (policy-controllable); noise with the same variance drags severely yet never locks (§5.3). The two axes are necessary in every *reward-channel* lock we observe (the 8B auxiliary-loss lock, §6, arises without either and sits outside this criterion's scope) but not jointly sufficient: the Δacc and anchor-QA arms sustain variance on policy-controllable signals and neither locks. What further separates the locking arms (plausibly, the existence of a degenerate behavior that maximizes the signal at zero task cost, as a fully predictable loop does for difference-form prediction) remains uncharacterized. One more ε-scale term shares the pool: the invalid-action penalty (0.1, the same order as λ), which varies per step-sample even inside an all-fail group. Strictly, the amplifier stretches the *joint* spread of shaping and penalty, rewarding predictability and surface compliance together, precisely the pair the collapsed policy ends up maximizing (invalid rate 0.005, prediction-parse validity 1.00, §5). Proposition 1 states both limits explicitly, and the measured three-point sweep shows no λ trend, consistent with either. The sweep alone therefore cannot apportion the pooled spread between the two terms (penalty-dominated pools are λ-invariant precisely because the shaping is negligible there). The decomposition is *implementable* as a training-time monitor, since the two terms enter the reward separately; our runs logged only their batch means, so we do not report the share, and the apportioning we rely on is the causal one below. Every baseline arm carries the same penalty under the same normalizer and the same early all-fail exposure and never collapses, so penalty spread alone does not produce the failure; adding the shaping term does (the knock-out matrix, §5.3), and the terminal behavior is prediction-optimal (accuracy driven to 1.0), the imprint of shaping pressure that dominated during the approach (at saturation itself the shaping contribution has vanished, Proposition 2).

The criterion is also consistent with the published reward-channel successes of §2, in the weak sense that neither success contradicts it. RWML's embedding-space gap reward is level-scored, outside the difference-form family whose per-step fluctuation feeds the amplifier; for level signals our own pair shows the outcome hinges on whether within-group spread decays (self-report) or persists (anchor-QA), a property we cannot measure in RWML from outside, and its authors' warning that token-level next-state prediction "can lead to model collapse" independently marks the corner we map. VAGEN redesigns the estimator, and our GiGPO arm shows redesign alone does not decide (the deciding bit is whether the within-group σ division survives, a coordinate VAGEN's description leaves unreported). The criterion issues no safety certificates here; it is compatible with these successes, not confirmed by them.